Aerothermodynamic Analyses for the LOFTID Technology Demonstration Mission

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On November 10, 2023, the LOFTID flight test successfully demonstrated the aerodynamic and thermal protection system performance of an inflatable aeroshell at conditions relevant to an operational mission. Aerodynamic performance and aeroheating environment databases for this mission were generated using multiple computational tools for the rarefied, hypersonic, and supersonic flow regimes, supplemented by wind tunnel testing to obtain aeroshell boundary-layer transition and wake flow simulation validation data. A detailed discussion of tools, methods and results is presented herein.

I. Nomenclature

C_A	axial force coefficient
$C_{m,CG}$	pitching moment coefficient about center-of-gravity
C_N	normal force coefficient
D	vehicle diameter (m)
H_0	total enthalpy (J/kg)
H_{AW}	adiabatic wall enthalpy (J/kg)
h	heat-transfer film coefficient (kg/m^2-s)
h_{FR}	Fay-Riddell theory heat-transfer film coefficient
Kn_{D}	Knudsen number based on diameter
M_∞	free stream Mach number
p_{w}	surface wall pressure (Pa)
q_{conv}	convective heat transfer rate (W/cm^2)
q_{rad}	radiative heat transfer rate (W/cm^2)
Q_{conv}	convective heat transfer load (kJ/cm^2)
Q_{rad}	radiative heat transfer load (kJ/cm^2)
R_C	corner radius (m)
R_N	nose radius (m)
Re_∞	free stream unit Reynolds number (1/m or 1/ft)
Re_θ	momentum thickness Reynolds number
T_∞	free stream temperature (K)
U_∞	free stream velocity (m/s)
α	angle of attack (deg)
$\tau_{\mathrm{w,total}}$	wall total shear stress (Pa)
ρ_∞	free stream density (kg/m^3)

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II. Introduction and Background

A. Hypersonic Inflatable Aerodynamic Decelerator Technology Concept

Hypersonic Inflatable Aerodynamic Decelerator (HIAD) technology [1] offers an effective, mass-saving alternative to the conventional aerodynamic decelerators used in previous entry, descent, and landing systems such as the Orion crew capsule or the Mars Science Laboratory / Mars 2020 aeroshells. A HIAD has a lower mass and a smaller volume when packaged and stowed relative to a rigid aeroshell, but when deployed and inflated it provides a large surface area for drag production, resulting in a lower ballistic coefficient than a rigid aeroshell of equivalent surface area. The lower ballistic coefficient of the HIAD results in a less stressful trajectory with lower heat fluxes and integrated heat loads, thus reducing the performance demands on the aeroshell flexible thermal protection system (FTPS). For these reasons, HIAD systems are leading candidates for crew and cargo missions to Mars [2–4] that will require the delivery of landed masses up to a factor of 10 greater than has been possible using the rigid aeroshell technology of previous missions. HIAD systems are also under consideration for the return of payloads from Low Earth Orbit (LEO) or for recovery and reuse of launch vehicle components [5].

The inflatable decelerator concept was first proposed in the 1960s [6–7] and since then various configurations, such as ram-inflated [8], tension-cone [9], trailing ballute [10], etc., have been proposed and studied. However, no successful hypersonic flight test of an inflatable aeroshell had ever been conducted. Thus, NASA committed to a program of development and demonstration flights to mature and validate this technology beginning with the Inflatable Reentry Vehicle Experiment (IRVE) series of flight tests launched on sounding rockets from the NASA Wallops Flight Facility.

B. IRVE Series Suborbital Demonstration Flights

The first IRVE mission [11–13] suffered a launch system failure that was not related to the HIAD payload. A duplicate of the IRVE vehicle was built and in 2009 the IRVE-II [14–16] flight demonstrated exoatmospheric inflation of the aeroshell and aerodynamic stability at hypersonic conditions. Because the IRVE-II vehicle was launched from a small, two-stage Black Brant IX sounding rocket, the aerothermal environment was quite benign: the maximum Mach number was 6.2, which produced peak heating rates of only ~2W/cm^2. Nevertheless, the success of IRVE-II provided the foundation for the next step in HIAD technology development, the IRVE-3 mission.

The IRVE-3 mission [17–18] was a direct follow-on to IRVE-II, with the overall goal of demonstrating HIAD performance and gathering flight data in a mission-relevant environment. The primary mission objective for IRVE-3 was to demonstrate FTPS survivability during the hypersonic segment of the flight at heating levels an order-of-magnitude higher than IRVE-II. A secondary mission goal was to demonstrate the performance of an active center-of-gravity system during the subsonic portion of the trajectory. Key upgrades for the IRVE-3 mission included: launch aboard a 3-stage sounding rocket to reach a peak Mach number of ~10 (as compared to Mach ~ 6 for IRVE-II); a larger entry vehicle mass (2 × greater than IRVE-II), a flight-relevant FTPS design; an attitude control system to maintain vehicle orientation; and an extensive instrumentation suite, including heat-flux and pressure gauges on the solid nose and in-depth thermocouples at multiple locations on the inflatable aeroshell. The IRVE-3 mission flew successfully in 2012, reaching a peak Mach number of 10 and demonstrating FTPS performance and survivability at a heating rate of ~ 20W/cm^2.

After the successful IRVE series of suborbital, sounding rocket demonstration flights, the next step in NASA development of HIAD technology was a mission-scale demonstration launched from a heavy lift vehicle to reach orbital velocity, the Low-earth Orbit Flight Test of an Inflatable Decelerator (LOFTID).

C. LOFTID Orbital Reentry Demonstration Mission

In November of 2023, the LOFTID vehicle [19-21] was carried aboard an Atlas V rocket as a secondary payload on the Joint Polar Satellite System-2 (JPSS-2) mission. The vehicle (Figure 1) was comprised of an inflatable aeroshell packaged in a cylindrical 1.3 m diameter payload section. The aeroshell structure consisted of six large and one small heavy-duty, inflatable toroids held together by multiple tension straps and covered with flexible thermal protection system materials. After deployment from the launch vehicle, pressurized tanks in the payload inflated the toroids to form a 6 m diameter, 70-deg sphere-cone geometry.

LOFTID reentered the atmosphere at a velocity of ~8 km/sec and reached peak heating levels of ~40 W/cm^2 at Mach 25. The mission successfully demonstrated the aerodynamic, aeroheating, thermal, and structural performance of a large HIAD at conditions relevant to a heavy down-mass operational mission at Earth or Mars. The vehicle splashed down (Figure 2) in the Pacific Ocean and was recovered intact (Figure 3) except for minor damage to the nose cap from impact and recovery operations.

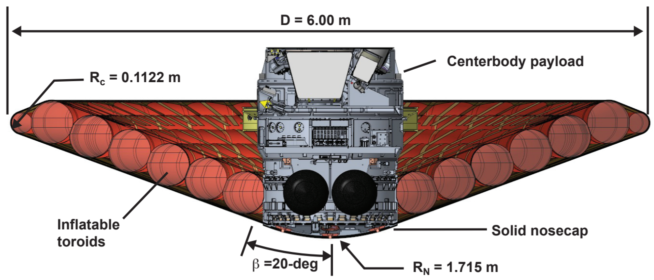


Figure 1. LOFTID flight vehicle cross section and dimension.

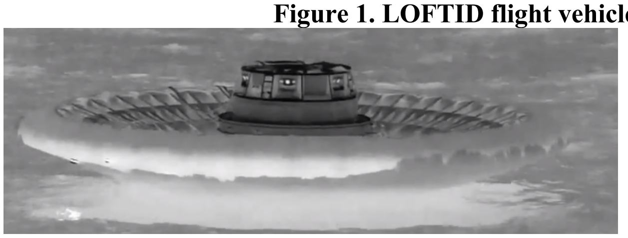


Figure 2. IR image of LOFTID splashdown [Source NASA].

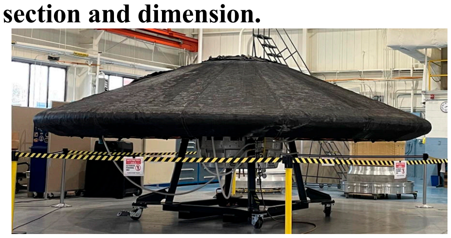


Figure 3. Recovered LOFTID flight vehicle [Source NASA]

LOFTID was heavily instrumented [22] to measure its performance during reentry. Aerothermodynamic instrumentation included heat-flux sensors, pressure gauges, and a radiometer on the nose section and thermocouples embedded at various depths in the FTPS at multiple locations on the front and back of the aeroshell. Additionally, visual and infrared cameras were mounted on the centerbody pointing forward to track the inflation, deployment, and in-flight deflections of the aeroshell structure along with its back-face temperature levels during reentry. Additional backward facing cameras were also mounted on the centerbody to provide visual imaging of the wake flow.

A key performance parameter of the mission was demonstrating the survivability at high temperatures of the multi-layer, flexible thermal protection system [23] covering the front face of the aeroshell. Unlike operational missions where design conservatism would be desirable to ensure safety and success, the margins and uncertainties in the aerothermodynamic environment on the front of the LOFTID aeroshell were minimized to ensure that the FTPS would experience relevant temperatures for validation of its thermal performance.

In contrast to the front-face of the aeroshell, the back-face carried no thermal protection because of the challenges in packaging and stowing FTPS material on the “inside” of the uninflated aeroshell within the payload. While the heating levels on the back of the aeroshell were roughly an order of magnitude lower than those on the front, the lack of FTPS meant that the inflatable toroids and tension straps were directly exposed to the wake flow field. Hence, extra conservatism was required in the definition of wake heating environments, which presented a design challenge to ensure high-heating on the front of the aeroshell while minimizing heating on the back.

III. Aerothermodynamic Tools and Methods

A combination of computational and experimental tools and methods were employed to develop the LOFTID flight aerothermodynamic database that defined the aerodynamic forces and moments from atmospheric entry to parachute deployment and the convective and radiative aeroheating fluxes and loads. The aerodynamic data were incorporated into six degrees-of-freedom Monte Carlo simulations to define the vehicle trajectory. The aeroheating data were employed as external boundary conditions for thermal analyses at selected points on the aeroshell rigid nose cap, the front and back of the inflatable aeroshell, and on the payload. Several design analysis cycles were performed to develop these databases. Additional post-flight analyses will also be conducted to assess the performance of the vehicle to evaluate the fidelity of the aerothermodynamic tools and methods used in the design of the vehicle.

A. Computational Methods

Multiple computational tools were employed for simulations along the LOFTID flight trajectory with overlapping application ranges to provide a complete database of aeroheating and aerodynamic information. Detailed computational results from the simulation data points were analyzed to develop curve-fitting / interpolation-extrapolation routines that were incorporated into trajectory simulations for rapid estimation of flight performance and for use in thermal analyses.

1. MAP Flow Field Solver

The Multi-physics Algorithm with Particles (MAP) is a Direct Simulation Monte Carlo (DSMC) Boltzmann solver for high-altitude, rarefied flow field simulations [24]. MAP employs path tracking of representative particle populations on Cartesian/cut-cell topology grids and includes reacting gas and surface catalysis models.

MAP was employed to determine the high-altitude aerodynamic and convective aeroheating environments for LOFTID from atmospheric entry down to Knudsen numbers of ~**O**(0.01). Additional hybrid LAURA continuum forebody / MAP rarefied wake solutions were performed down to lower altitudes to help define the aeroheating environments of the low-density wake flow.

2. LAURA Flow Field Solver

The Langley Aerothermodynamic Upwind Relaxation Algorithm (LAURA) is a three-dimensional, finite-volume structured grid topology Navier-Stokes solver for continuum flow field simulations [25, 26]. LAURA incorporates perfect-gas, equilibrium chemistry, and nonequilibrium reacting gas thermodynamic options, as well as a variety of turbulence models and coupled ablation and radiation transport models.

LAURA was employed to provide aerodynamic and convective aeroheating environments from the high-altitude hypersonic rarefied overlap with MAP down to the high supersonic (M ~ 4) overlap with FUN3D solutions. LAURA solutions were generated using a two-temperature (translational/rotational and vibroelectronic) 11-species reacting gas model (N_2, O_2, NO, N, O, N+, O+, NO+, N_2+, O_2+, e-) with a fully-catalytic, radiative equilibrium wall temperature boundary condition. Solutions were generated both for laminar flow and for turbulent flow using the algebraic Cebeci-Smith model. Although ionization along the trajectory is insignificant with respect to aerothermodynamic environments, the ionized species were included in the computations to provide the basis for supplemental evaluation of radio-frequency communications blackout.

3. FUN3D Flow Field Solver

The Fully Unstructured in Three Dimensions (FUN3D) code is a three-dimensional, finite-volume, unstructured grid topology Navier-Stokes solver for continuum super / trans / subsonic flows [27]. FUN3D incorporates perfect-gas and equilibrium air models and shares the nonequilibrium reacting gas thermodynamics models implemented in LAURA. FUN3D includes adjoint-based error estimation and grid adaption, which permit more efficient computations of complex vehicle topologies and flow fields.

The FUN3D solver was used to define aerodynamic performance from the low hypersonic overlap with LAURA down to subsonic conditions. FUN3D simulations were performed using a one-temperature, 5-species reacting gas model (N_2, O_2, NO, N, O) for hypersonic cases and a perfect-gas model for super / trans / subsonic aerodynamic cases.

4. HARA Radiation Transport Solver

The High-temperature Aerothermodynamic Radiation Algorithm is a radiation transport solver for evaluation of high-temperature shock-layer radiation in hypersonic flows [28, 29] and is directly coupled with LAURA to provide radiative heating rates for each simulation point. HARA includes non-Boltzmann population models for atomic and molecular radiation for all atmospheres of interest for planetary exploration missions. Radiation transport can be modelled through tangent-slab approximation or detailed ray-tracing and the code is integrated with LAURA to provide direct flow field / radiation transport coupling. For LOFTID simulations, the tangent-slab modeling was employed to determine aeroshell forebody radiative heating. Three-dimensional ray-tracing was performed for selected cases to evaluate aeroshell back-face and payload radiative heating, which was determined to be insignificant and was not included in the aeroheating database.

5. Computational Grid Topologies

Each of the computational tools requires a different grid generation methodology. For DSMC simulations of rarefied flow fields using MAP, a Cartesian grid topology with cut cells that conform to local surface features is used. The cut cells allow for accurate representation of local surface features, such as the exposed toroids on the back-face of the aeroshell. For super / trans / subsonic continuum flow field simulations using FUN3D, unstructured tetrahedral grids were used for the bulk of the flow field, while hexahedral elements were used to define the wall boundary-layer.

For hypersonic continuum simulations using LAURA, the flow field is usually defined by a multiblock, structured hexahedral grid topology. While such topologies provide highly accurate simulations over smooth surfaces with slowly-varying curvature, such as the front-face of an aeroshell, it can be extremely challenging to develop a structured-grid topology that can accurately resolve complicated geometries with regions of large curvature and convex surface features, such as the wake flow over the exposed toroids on the back-face of LOFTD. Furthermore, even a well-resolved initial wake flow grid is prone to becoming highly-skewed by the shock and boundary-layer grid adaptation process, producing artificial numerical dissipation, and hindering the solution convergence process.

For these reasons, a new "stacked-block" structured grid topology and boundary conditions and a new algorithm to adapt and smooth these grids have been implemented [30] in the LAURA code to permit more accurate resolution of complicated flow fields. The stacked-block grid topology removes the limitation of defining a grid with continuous *k*-index grid lines from the surface through the shock layer and into the freestream. Instead, the flow domain is decomposed into feature-resolving blocks with arbitrary (*i, j, k*) face-to-face orientation. As illustrated in Figure 4, three new stacked-block topology grid block types have been added to the original continuous *k*-line block type, which are defined as:

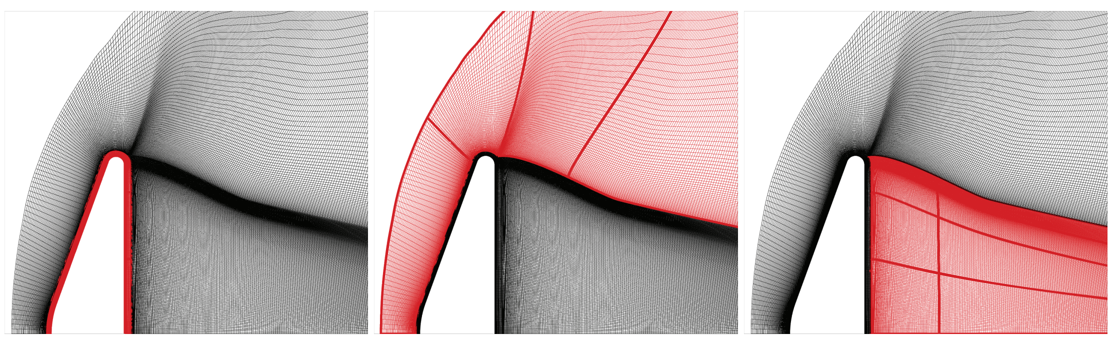
Type-1: Boundary-layer block with solid-surface *k*-min boundary condition interfacing with a Type-2 block or a Type-4 block at the *k*-max boundary. Type-1 blocks are adapted to cluster points closer to the wall at the *k*-min boundary, but their outer *k*-max boundary remains fixed.

Type-2: Outer-boundary block with a freestream *k*-max boundary interfacing with a Type-1 block or a Type-4 block at the *k*-min boundary. Type-2 blocks are adapted to expand/contract the *k*-max boundary to encompass the shock, but their inner *k*-min boundary remains fixed.

Type-3: Standard continuous *k*-line block which can be adapted to cluster cells near the wall and at the shock and to extend/contract the outer boundary to fully encompass the bow shock.

Type-4: Inner block which is bounded on all faces by other Type-1, -2 or -4 blocks. The *i-j-k* orientation of these blocks with respect to their neighbors is arbitrary. Type-4 blocks cannot be adapted, and their physical *x-y-z* coordinates remain fixed.

This stacked block grid topology allows for much greater flexibility in grid design and results in greater fidelity than the standard topology for such configurations LOFTID with the convex shape of the aeroshell back face, exposed toroids and recessed centerbody. An example of the stacked-block grid topology implementation for LOFTID wind tunnel case is shown in Figure 5.



Type-1: Boundary layer block　　Type-2 Outer boundary block　　Type-4 Interior block

Figure 4. Stacked-block grid topology block types.

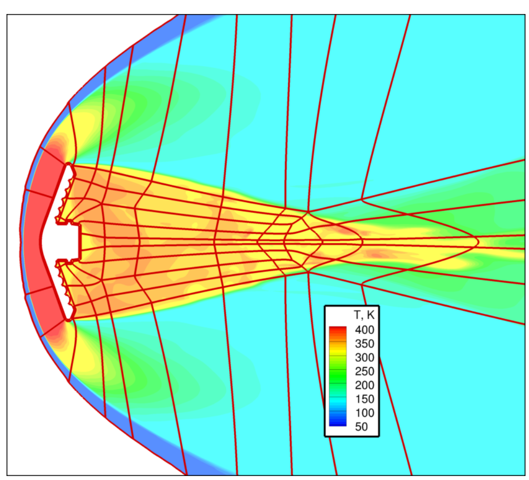


Figure 5. Stacked-block grid topology for LOFTID at wind tunnel conditions.

B. Experimental Methods

Wind tunnel tests were conducted to obtain experimental data on physical phenomena for which computational methods are not suitable or have not been validated. Boundary-layer transition and turbulent heating augmentation due to the deflection of the aeroshell FTPS from a smooth outer mold line (OML) are one such fluid dynamics/aeroheating problem that are beyond the state-of-the-art of the computational methodology. Multiple wind tunnel tests were performed to generate experimental data on these phenomena. Another computational challenge was the prediction of the LOFTID wake structure, which is defined by a large, separated and recirculating, unsteady, and potentially turbulent flow field. While computational predictions for the wake flow are part of the LOFTID database, the uncertainty in these computations was large. Thus, experimental data on the wake flow environment were obtained through multiple wind tunnel tests to help define the wake flow structure and reduce the computational uncertainty levels.

1. NASA Langley Aerothermodynamics Laboratory Wind Tunnels

The NASA Langley Aerothermodynamics Laboratories (LAL) is comprised of three conventional, blow-down to vacuum hypersonic wind tunnels as well as a large vacuum sphere test chamber [31]. Two of the LAL wind tunnels, the 31-Inch Mach 10 Air Tunnel and the 20-Inch Mach 6 Air Tunnel, were utilized in the LOFTID project.

The NASA LaRC 31-Inch Mach 10 Air Tunnel is a perfect-gas, blow-down facility in which dried, filtered air is used as the working fluid. The tunnel has been calibrated for reservoir conditions varying from 150 psi to 1450 psi at an operating temperature of 1850°R, which produces free stream unit Reynolds numbers from 0.25×10^6/ft to 2.2×10^6/ft. The nozzle is water-cooled, has a three-dimensional contour, and ends with a 1.07×1.07 in square throat. The 31×31 in square test section features optical access through side, top, and bottom windows for visual imaging techniques and has a side-mounted injection system with a ±45 deg pitch range and ±5 deg yaw range. The test core varies from approximately 12×12 in at the lowest unit Reynolds number to 14×14 in at the highest unit Reynolds number. This tunnel has the highest Mach number of the LAL facilities and is mainly employed in aerodynamic and fluid-mechanics studies. The high stagnation temperature of the facility also makes it suitable for aftbody/wake heating studies where the temperature rise is much lower than on the forebody.

The NASA LaRC 20-Inch Mach 6 Air Tunnel is a blow-down facility in which heated, dried, and filtered air is used as the test gas. The tunnel has a two-dimensional contoured nozzle that opens into a 20.5×20.0 in test section. The tunnel is equipped with a bottom-mounted injection system with a –5 deg to +55 deg pitch range and ±5 deg yaw range that can transfer a model from a sheltered model box to the tunnel centerline in less than 0.5 sec. Run times of up to 15 minutes are possible in this facility, although for the current aeroheating study run times of only a few seconds were required. The nominal reservoir conditions of this facility produce perfect-gas free stream flows with Mach numbers between 5.8 and 6.1 and unit Reynolds numbers of 0.5×10^6/ft to 8.3×10^6/ft. With the wide Reynolds number operating range capable of producing laminar, transitional, or turbulent flow on most geometries, this tunnel is primarily used for heat-transfer and boundary-layer transition studies.

2. Instrumentation Techniques

Phosphor thermography was employed to obtain aeroheating data for the LOFTID project. Phosphor thermography is an optical technique for measurement of global surface temperatures on wind tunnel test models, from which heat-transfer rates and boundary-layer transition locations can be determined [32, 33]. Phosphor thermography has been employed in the LAL facilities for over 20 years and provides quantitative aeroheating data.

In addition to the heat transfer measurements, test technique development studies [36] were conducted using the LOFTID configuration to obtain flow field visualization data using nitric-oxide based Planar Laser-Induced Fluorescence (NO-PLIF) and Femtosecond Laser Electronic Excitation Tagging (FLEET). The purpose of these studies was to demonstrate the capability for wake flow field characterization using nonintrusive, laser-based diagnostics.

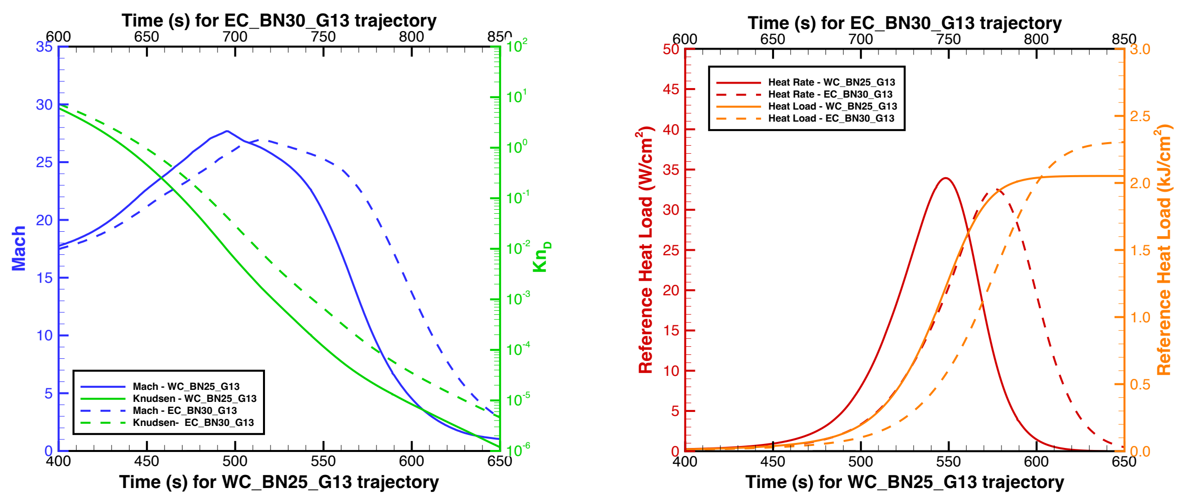
IV. Aerothermodynamic Environments

A. Trajectory Parameters

Aerodynamic performance and aeroheating environments databases were developed for several iterations of the LOFTID trajectory over the course of the project. Trajectory parameters (Mach and Knudsen numbers and reference heating rate and heat load) for two of these – the *EC_A_BN30_G13* and *WC_BN25_G13* – are shown in Figure 6 and are listed in Table 1 and Table 2. An initial aeroheating database was generated along the *EC_A_BN30_G13* trajectory and subsequent updates to the aeroshell backface and centerbody heating databases were all generated using this trajectory to evaluate wake flow computational methodology improvements over the course of the project. The final aerodynamic performance and aeroshell forebody heating databases were generated using the *WC_BN25_G13* trajectory.

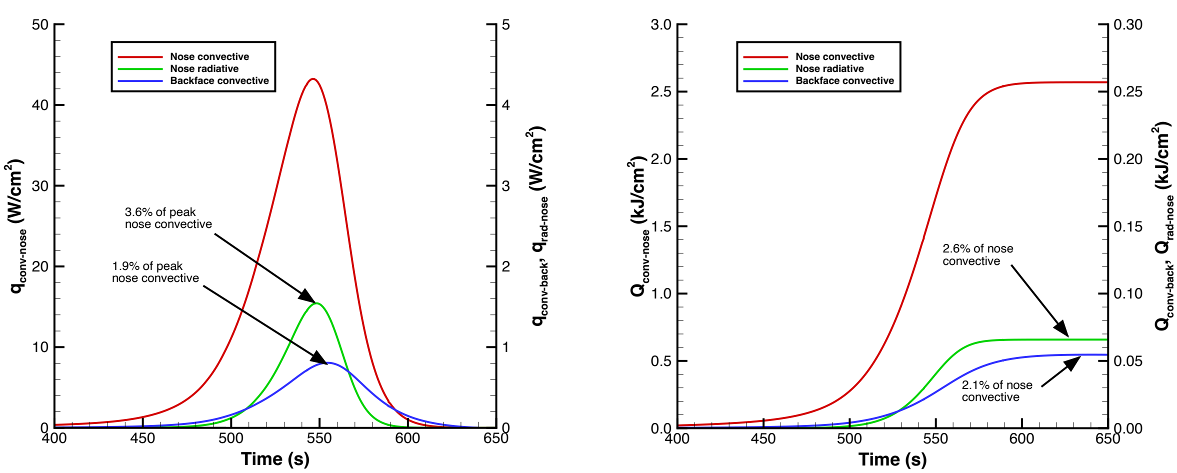
The nominal angle-of-attack for LOFTID was 0 deg, with expected deviations of less than 10 deg from atmospheric entry through supersonic conditions. To encompass this angle-of-attack range, computational solutions were generated for $\alpha = 0$ deg to 20 deg. Each of the three solvers, MAP, LAURA and FUN3D were employed over different Mach-Knudsen number ranges, as appropriate, with overlap between the ranges as illustrated in Figure 8. Rarefied flow field solutions were generated using MAP for a Knudsen number range of $Kn_D = \mathbf{O}(10^{+1})$ to $\mathbf{O}(10^{-3})$, which bounds atmospheric entry to the begin of the heat pulse. Hypersonic/supersonic continuum flow field solutions were generated using LAURA from $Kn_D = \mathbf{O}(10^{-2})$ at high altitude down to $\boldsymbol{M_\infty} \sim 4$ at low altitude. Super / trans / subsonic continuum solutions were generated using FUN3D from $\boldsymbol{M_\infty} \sim 6$ down to $\boldsymbol{M_\infty} \sim 0$.

The nose convective and radiative and back-face peak convective rates and loads from the aeroheating database are shown in Figure 7. Notably, the radiative heating on the front-face of the aeroshell and the convective heating rates and loads on the back-face of the aeroshell were very small (less than 5% of convective) with respect to the nose convective levels. The radiative heating component had minimal effect on the design and analysis of LOFTID. However, significant effort was spent to characterize the wake flow environment because the back-face of the aeroshell had no FTPS, leaving the toroids exposed directly to heating.



a) Mach and Knudsen numbers b) Reference hemisphere convective heating

Figure 6. Representative preflight trajectory parameters.



a) Heating rates b) Heating loads

Figure 7. Key heating parameters along the WC_BN25_G13 trajectory.

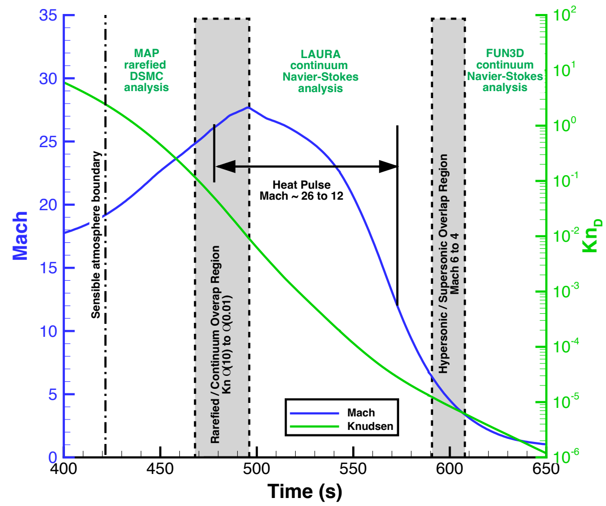


Figure 8. Computational tools overlap across the Mach / Knudsen number range.

Table 1. EC_BN30_G13 Trajectory Parameters for wake flow environments.

t **(s)**	***Alt.*** **(km)**	U_∞ **(m/s)**	ρ_∞ **(kg/m³)**	T_∞ **(K)**	M_∞	Re_D	Kn_D	q_∞ **(Pa)**	**Solver**
546	159.76	7301.7	4.0941E-10	525.1	15.90	6.4821E-01	3.1377E+01	1.0914E-02	MAP
600	140.58	7325.7	1.7427E-09	437.0	17.48	3.1413E+00	7.3714E+00	4.6761E-02	MAP
650	121.51	7349.4	1.3537E-08	300.6	21.15	3.2230E+01	9.4899E-01	3.6558E-01	MAP
685	107.45	7365.6	1.3164E-07	223.3	24.59	3.9851E+02	9.7582E-02	3.5709E+00	MAP
712	96.22	7368.8	1.2049E-06	187.3	26.86	4.2386E+03	1.0661E-02	3.2713E+01	MAP, LAURA
721	92.40	7360.8	2.5013E-06	188.5	26.75	8.7391E+03	5.1358E-03	6.7761E+01	MAP, LAURA
728	89.42	7346.3	4.2691E-06	192.2	26.43	1.4634E+04	3.0091E-03	1.1520E+02	LAURA
738	85.12	7303.9	8.6901E-06	197.2	25.95	2.8972E+04	1.4782E-03	2.3179E+02	MAP, LAURA
749	80.37	7203.9	1.8008E-05	201.2	25.34	5.8212E+04	7.1335E-04	4.6728E+02	MAP, LAURA
753	78.64	7145.3	2.3670E-05	202.6	25.04	7.5443E+04	5.4271E-04	6.0423E+02	LAURA
758	76.47	7046.4	3.3369E-05	204.2	24.60	1.0417E+05	3.8497E-04	8.2840E+02	LAURA
767	72.59	6770.4	6.0561E-05	210.2	23.29	1.7723E+05	2.1212E-04	1.3880E+03	LAURA
776	68.77	6327.4	1.0531E-04	219.4	21.31	2.7791E+05	1.2198E-04	2.1082E+03	LAURA
782	66.27	5927.8	1.4742E-04	227.1	19.62	3.5406E+05	8.7137E-05	2.5901E+03	LAURA
784	65.45	5776.7	1.6424E-04	229.7	19.01	3.8085E+05	7.8214E-05	2.7404E+03	LAURA
786	64.63	5617.2	1.8270E-04	232.2	18.39	4.0822E+05	7.0313E-05	2.8824E+03	LAURA
788	63.82	5450.0	2.0292E-04	234.8	17.74	4.3601E+05	6.3305E-05	3.0136E+03	LAURA
790	63.02	5275.7	2.2492E-04	237.3	17.09	4.6375E+05	5.7115E-05	3.1301E+03	LAURA
795	61.05	4815.5	2.8828E-04	243.5	15.40	5.3132E+05	4.4561E-05	3.3425E+03	LAURA
799	59.52	4431.4	3.4861E-04	248.1	14.04	5.8236E+05	3.6849E-05	3.4230E+03	LAURA
805	57.28	3850.0	4.5808E-04	254.0	12.05	6.5215E+05	2.8043E-05	3.3950E+03	LAURA
812	54.77	3199.2	6.1844E-04	260.4	9.89	7.1718E+05	2.0772E-05	3.1649E+03	LAURA
817	53.04	2769.8	7.6451E-04	262.9	8.52	7.6178E+05	1.6803E-05	2.9327E+03	LAURA
822	51.36	2378.8	9.3733E-04	265.3	7.29	7.9636E+05	1.3705E-05	2.6520E+03	LAURA
832	48.15	1722.2	1.3959E-03	267.0	5.26	8.5440E+05	9.2027E-06	2.0701E+03	LAURA
842	45.11	1225.1	2.0523E-03	266.5	3.74	8.9478E+05	6.2594E-06	1.5401E+03	LAURA

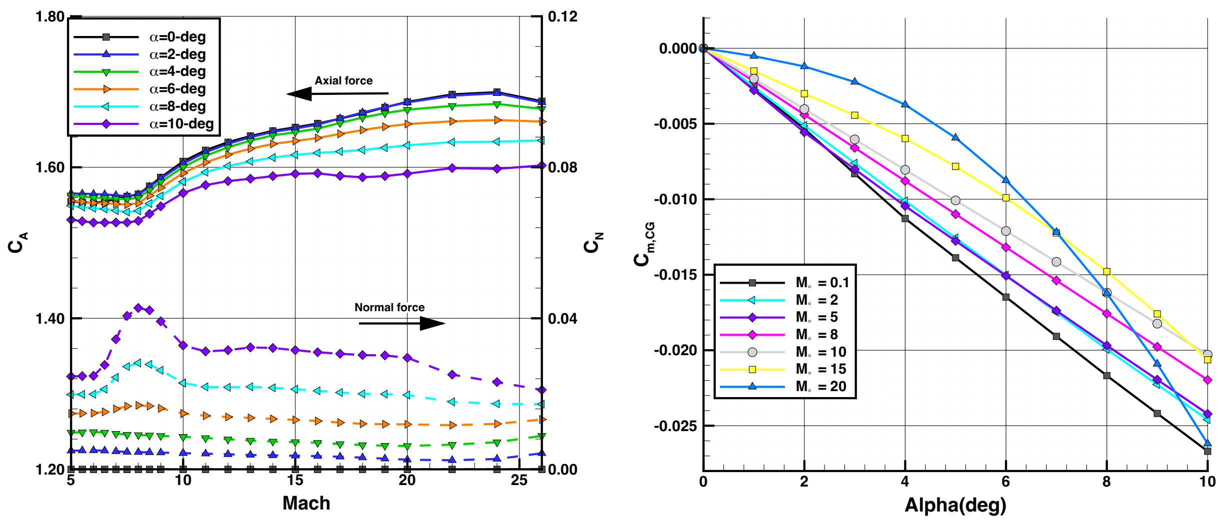
Table 2. WC_BN25_G13 Trajectory Parameters for aerodynamics and forebody aeroheating

t (s)	$Alt.$ (km)	U_∞ (m/s)	ρ_∞ (kg/m^3)	T_∞ (K)	M_∞	Re_D	Kn_D	q_∞ (Pa)	Solver
417	133.36	7584.5	4.2330E-09	343.6	18.80	9.4503E+00	3.0181E+00	1.2175E-01	MAP
438	123.57	7596.5	1.2809E-08	288.1	21.04	3.2751E+01	9.9737E-01	3.6960E-01	MAP
456	114.90	7606.8	4.3039E-08	242.7	23.39	1.2629E+02	2.9684E-01	1.2452E+00	MAP
471	107.48	7614.7	1.4026E-07	216.2	25.20	4.5336E+02	9.1087E-02	4.0663E+00	MAP
488	98.85	7619.6	6.5285E-07	191.7	27.20	2.3394E+03	1.9569E-02	1.8952E+01	LAURA
488	98.85	7619.6	6.5285E-07	191.7	27.20	2.3394E+03	1.9569E-02	1.8952E+01	MAP
494	95.76	7617.8	1.1600E-06	187.5	27.61	4.2366E+03	1.1013E-02	3.3659E+01	LAURA
501	92.11	7609.7	2.2326E-06	194.2	27.20	7.9029E+03	5.7223E-03	6.4643E+01	MAP, LAURA
515	84.74	7550.9	7.6252E-06	206.4	26.22	2.5416E+04	1.6754E-03	2.1738E+02	MAP, LAURA
525	79.42	7433.6	1.6826E-05	214.8	25.30	5.3388E+04	7.5928E-04	4.6488E+02	MAP, LAURA
537	73.05	7106.7	4.2284E-05	223.2	23.73	1.2420E+05	3.0214E-04	1.0678E+03	MAP, LAURA
546	68.36	6623.6	8.2743E-05	229.8	21.79	2.2111E+05	1.5440E-04	1.8150E+03	LAURA
551	65.83	6235.4	1.1807E-04	234.2	20.33	2.9243E+05	1.0821E-04	2.2952E+03	LAURA
559	61.97	5436.2	2.0191E-04	238.7	17.56	4.2925E+05	6.3274E-05	2.9835E+03	LAURA
563	60.13	4973.0	2.5941E-04	240.5	16.01	5.0135E+05	4.9250E-05	3.2077E+03	LAURA
569	57.53	4243.8	3.6673E-04	245.2	13.53	5.9548E+05	3.4837E-05	3.3024E+03	LAURA
578	53.97	3202.7	5.8222E-04	251.1	10.08	6.9984E+05	2.1943E-05	2.9861E+03	LAURA, FUN3D
585	51.45	2516.7	8.0437E-04	254.2	7.88	7.5221E+05	1.5883E-05	2.5474E+03	LAURA, FUN3D
592	49.11	1952.8	1.0890E-03	255.4	6.10	7.8722E+05	1.1731E-05	2.0764E+03	LAURA, FUN3D
598	47.23	1558.8	1.4027E-03	254.2	4.88	8.1265E+05	9.1076E-06	1.7042E+03	LAURA, FUN3D
605	45.15	1194.5	1.8560E-03	252.8	3.75	8.2767E+05	6.8834E-06	1.3242E+03	LAURA, FUN3D
611	43.45	951.3	2.3689E-03	248.9	3.01	8.5176E+05	5.3932E-06	1.0719E+03	LAURA, FUN3D
622	40.49	632.9	3.6599E-03	241.8	2.03	8.9642E+05	3.4907E-06	7.3303E+02	FUN3D
631	38.19	468.4	5.1722E-03	236.5	1.52	9.5450E+05	2.4700E-06	5.6730E+02	FUN3D
651	33.19	305.6	1.1194E-02	226.8	1.01	1.3953E+06	1.1412E-06	5.2281E+02	FUN3D

A. Aerodynamic Performance

An aerodynamic performance database was developed for the LOFTID mission and incorporated into the Program to Simulate Optimized Trajectory (POST) six degree-of-freedom trajectory analysis tool [39]. This database includes inputs generated from LOFTID-specific simulations using the computational tools describe above, free-molecular regime simulations performed for the IRVE-3 vehicle, and subsonic dynamic stability data obtained from ballistic range and low-speed wind tunnel testing during the IRVE-3 program. The database was formulated in terms of aerodynamic coefficient dependency on either Mach number or Knudsen number, depending on the flight regime.

Super / hyper sonic aerodynamic performance parameters from the database are given here with the normal force and axial force coefficients shown in Figure 9(a) and the pitching moment coefficient in Figure 9(b). A comprehensive discussion of the LOFTID aerodynamic database and flight performance across the speed range is presented in Refs. [40-42]. The accuracy of the aerodynamic database resulted in the actual splashdown point within 3 nautical miles from the pre-flight prediction.



c) Axial and normal force coefficients d) Pitching moment coefficient

Figure 9. Super / hypersonic aerodynamic coefficients.

B. Aeroheating

An aeroheating database for the LOFTID mission was developed through computational analyses using the LAURA code for the continuum hypersonic regime and the MAP code for the rarefied flow regime. The computational database was supplemented by testing in the NASA LARC 31-Inch Mach 10 Air and 20-Inch Mach 6 Air Tunnels, which provided data on boundary-layer transition, turbulent heating augmentation due to OML scalloping, and aeroshell back-face data for verification of computed heating levels. This database was incorporated into the POST trajectory simulation tool to provide trajectory-based heating values for use in the thermal analysis of the vehicle FTPS. Database output of convective heating environments was formulated in terms of the film coefficient, *h* defined from the radiative equilibrium wall heating and enthalpy difference term as per Eq. (1), to remove the effects of wall temperature from the analysis, although dimensional flux values will be shown herein for convenience. Radiative heating output was formulated in terms of the dimensional flux since flow field radiation is not affected by wall temperature, and the effects of black-body radiative emissions from the wall to the flow field are dealt with in the thermal analysis.

For implantation in the flight database, curve-fits to a surface quantity of interest *Y* were generated from the solution set in terms of free stream density and velocity as per Eq. (2). This formulation is (relatively) trajectory independent, allowing for rapid updates to environments as trajectories change without the requirement of regenerating high-fidelity flow solutions. Angle-of-attack dependency was incorporated by fitting the *A*, *B*, *C* coefficients to the predictions in terms of angle-of attack as per Eq. (3) Fits were generated for multiple points of interest for thermal analysis including the geometric nose, the interface between the solid nose cap and inflatable aeroshell conical section; the peak location at the shoulder of the aeroshell; and the maximum heating location on the back-face of the aeroshell.

$$\boldsymbol{h} = q_{\boldsymbol{conv}}/(H_{\boldsymbol{0}} - H_W) \tag{1}$$

$$\boldsymbol{Y} = \boldsymbol{A} \times (\rho_\infty)^B \times (U_\infty)^C \tag{2}$$

$$\begin{aligned} \boldsymbol{A} &= \boldsymbol{A_1} + \boldsymbol{A_2} \times (\propto) + \boldsymbol{A_3} \times (\propto)^{\boldsymbol{2}} \\ \boldsymbol{B} &= \boldsymbol{B_1} + \boldsymbol{B_2} \times (\propto) + \boldsymbol{B_3} \times (\propto)^{\boldsymbol{2}} \\ \boldsymbol{C} &= \boldsymbol{C_1} + \boldsymbol{C_2} \times (\propto) + \boldsymbol{C_3} \times (\propto)^{\boldsymbol{2}} \end{aligned} \tag{3}$$

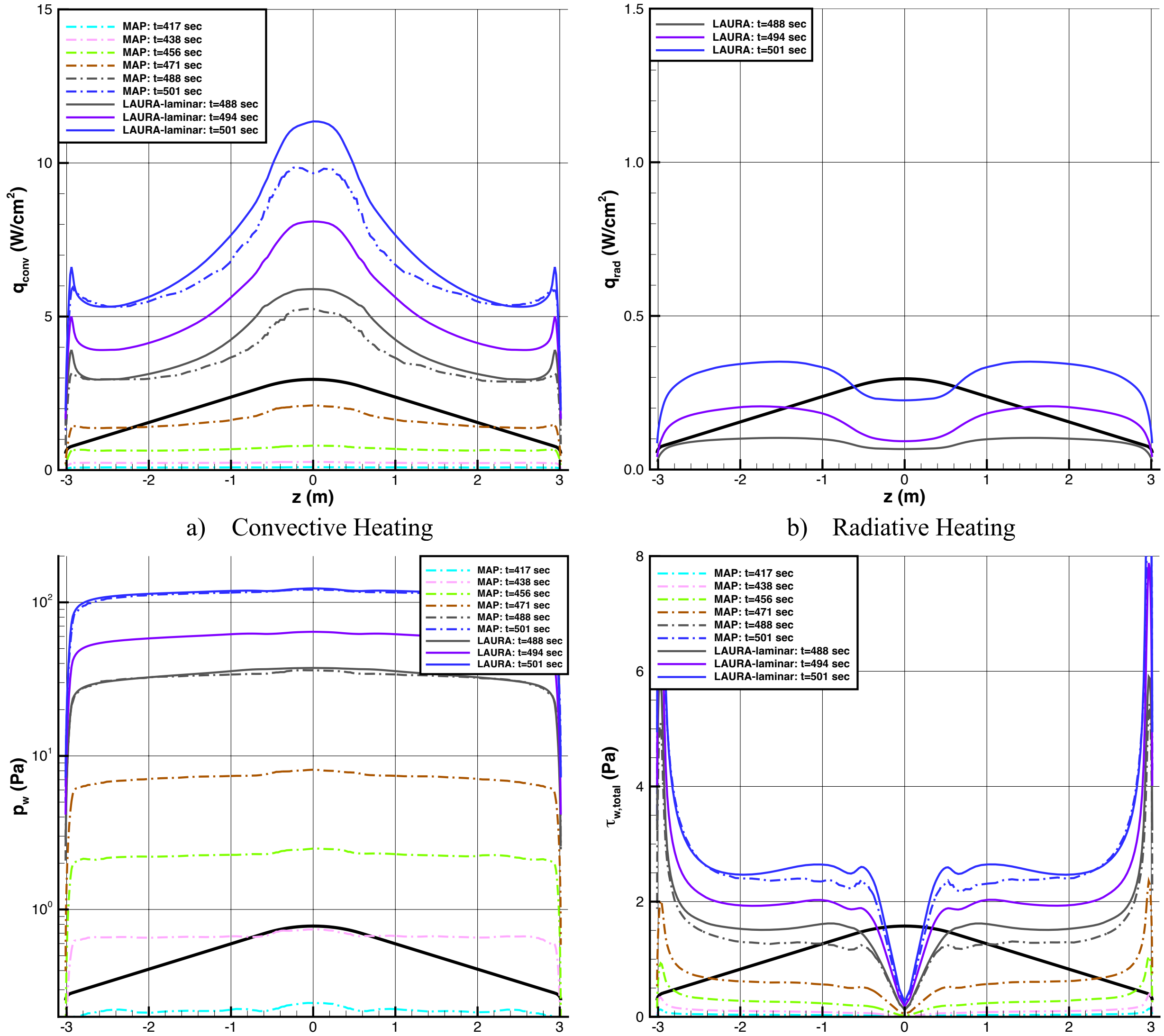
1. Aeroshell Front-Face Environments

Aerothermodynamic environments for the front-face of the LOFTID aeroshell are based on laminar and turbulent continuum LAURA simulations and MAP rarefied flow simulations at the WC_BN25_G13 trajectory points listed in Table 2. Solutions were generated at α = 0, 2, 4 and 8 deg to cover the potential range of deviations from the nominal flight value α = 0 deg. The simulations were all performed on a representative smooth OML geometry that did not incorporate the backward facing step at the nose / cone junction or any deformation of the FTPS. Convective and radiative heating, surface pressure, and total shear stress distributions for the front-face of the aeroshell are shown for three separate time increments defining: the rarefied flow regime in Figure 10; the high hypersonic flow regime in Figure 11, and the low hypersonic to supersonic regime in Figure 12.

Good agreement between LAURA and MAP predictions was achieved in the overlap region where both solvers were employed. Owing to the relatively low Reynolds numbers along the trajectory, the differences in predicted heating between laminar and turbulent LAURA solutions were not large, although the differences did increase for higher angle-of-attack cases.

The influence of angle of attack on these aerothermodynamic quantities is illustrated for two flight conditions in Figure 13 for a rarefied/continuum overlap case and in Figure 14 for the peak dynamic pressure case. For these small differences in angle of attack, the variation in surface quantities were not large, although the increase in turbulent heating on the leeside of the aeroshell (positive x-direction) is notable for the peak dynamic pressure case.

Although both laminar and turbulent LAURA solutions were generated, only the turbulent heating predictions were incorporated into the flight database. Even though a purely smooth-OML boundary-layer transition assessment indicated laminar flow, as shown by the momentum thickness Reynolds number information in Figure 15, it was assumed that the "as-built" vehicle would be exposed to turbulent flow. In contrast to the smooth-OML employed in the simulations, on the actual vehicle the influence of the backward facing step at the nose / cone junction, FTPS deformations due to aerodynamic loading, as well as the inherent roughness of the woven FTPS surface, would all act to produce turbulent flow. As will be shown later, the turbulent predictions bounded the effects of the FTPS deformation effects on heating observed in the wind tunnel data and it was assumed the same would be true for flight conditions.



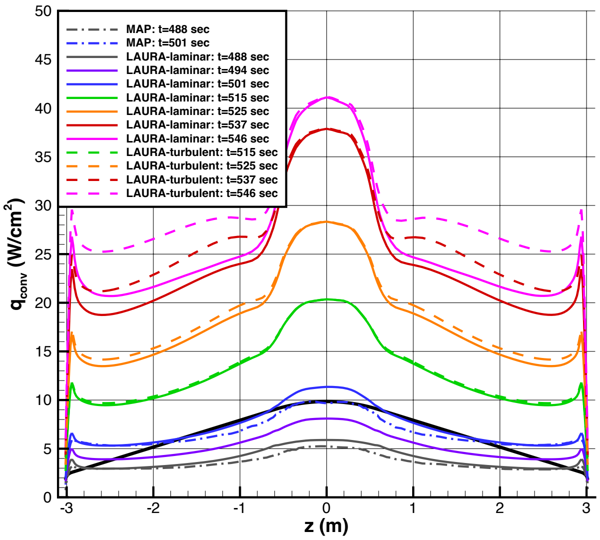
a) Convective Heating

b) Radiative Heating

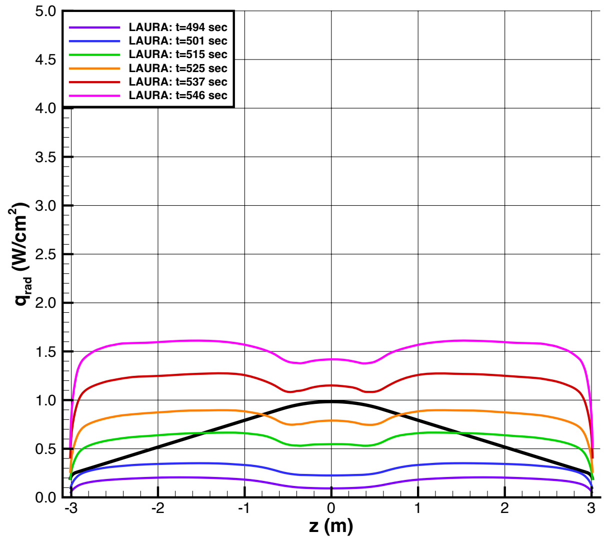
c) Surface Pressure

d) Surface Shear

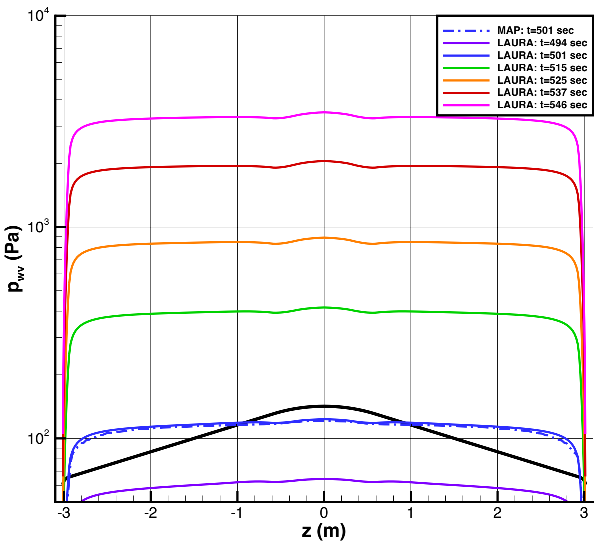
Figure 10. Aerothermodynamic parameters along symmetry plane, t = 417 s to 501 s.



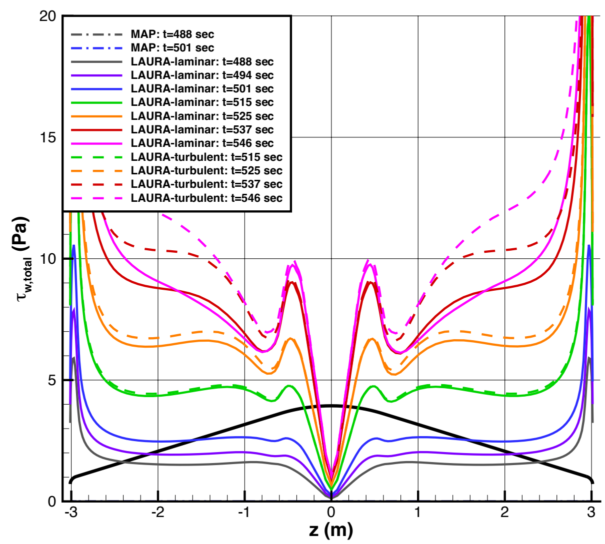
a) Convective Heating



b) Radiative Heating

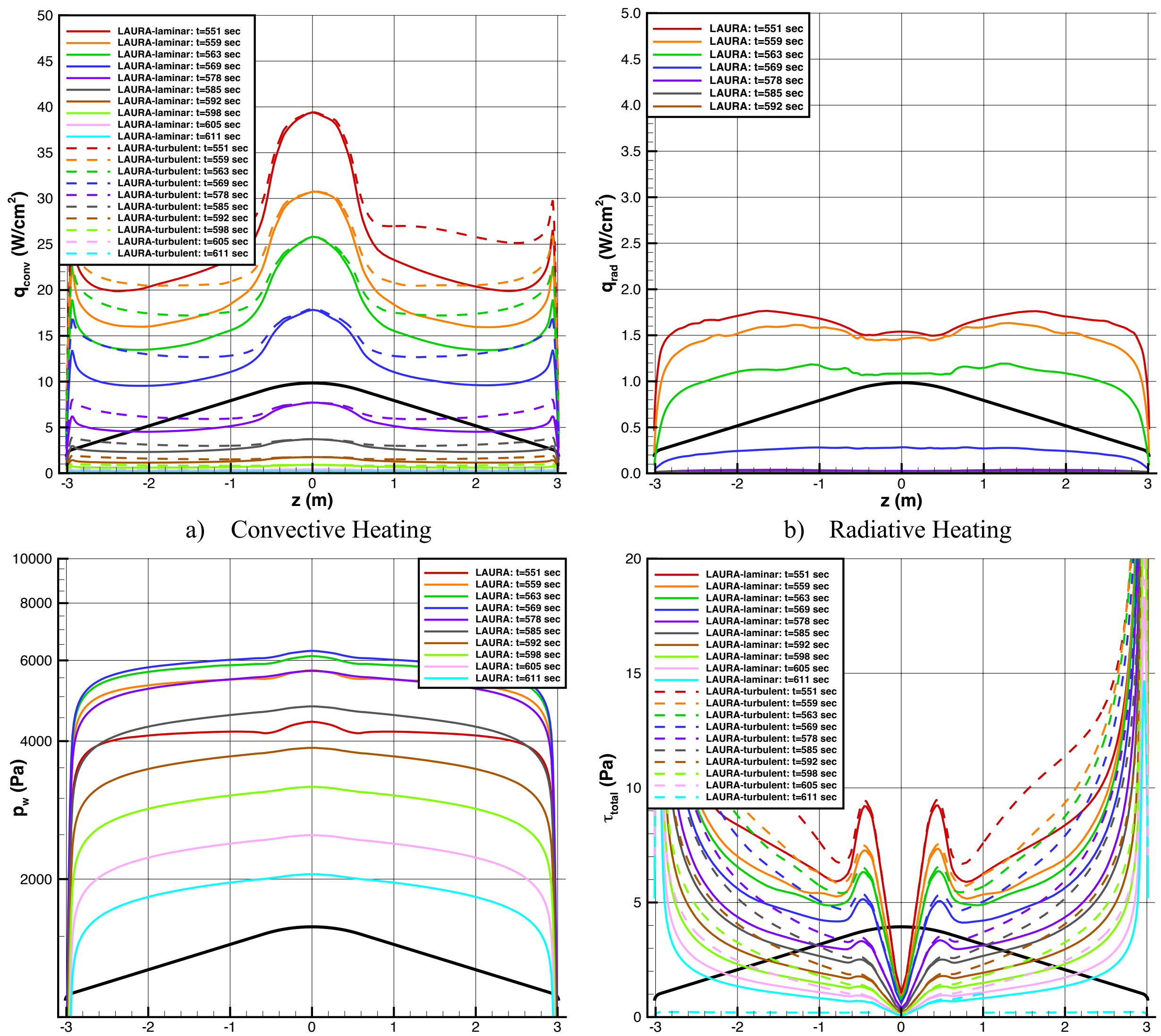


c) Surface Pressure



d) Surface Shear

Figure 11. Aerothermodynamic parameters along symmetry plane, t = 488 s to 546 s.



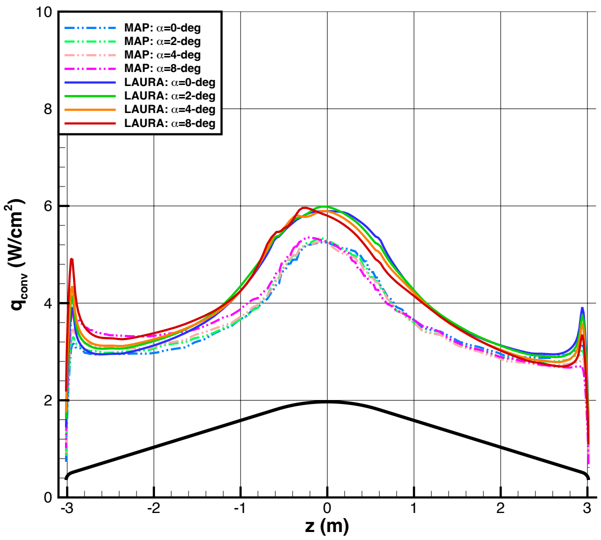
a) Convective Heating

b) Radiative Heating

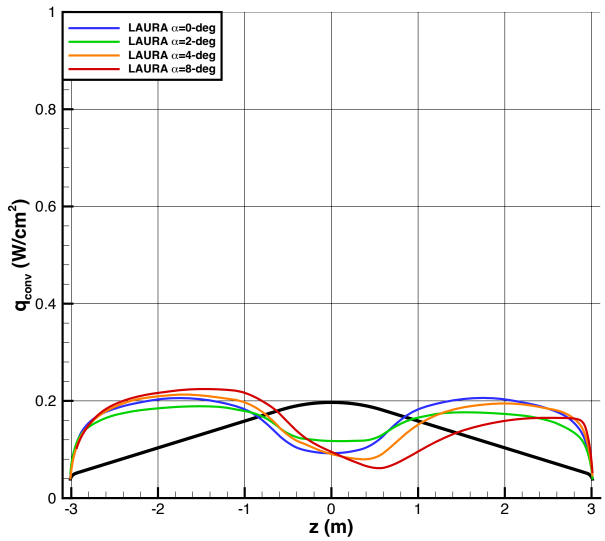
c) Surface Pressure

d) Surface Shear

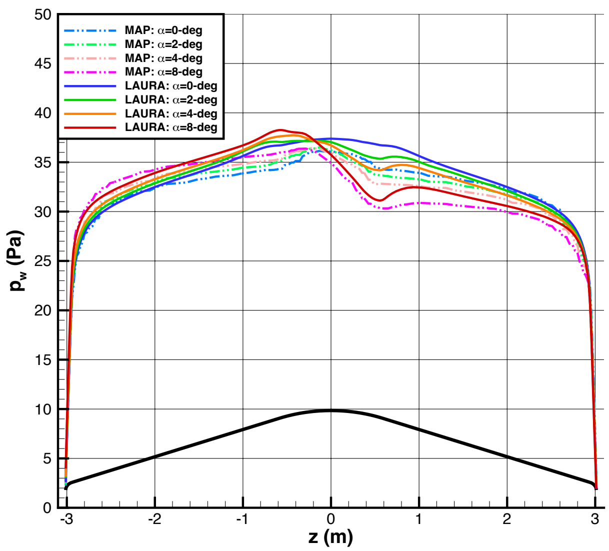
Figure 12. Aerothermodynamic parameters along symmetry plane, t = 551 s to 611 s.



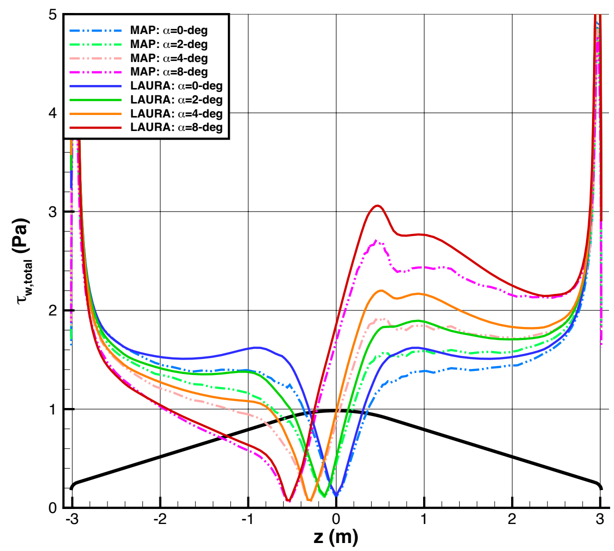
a) Convective Heating



b) Radiative Heating

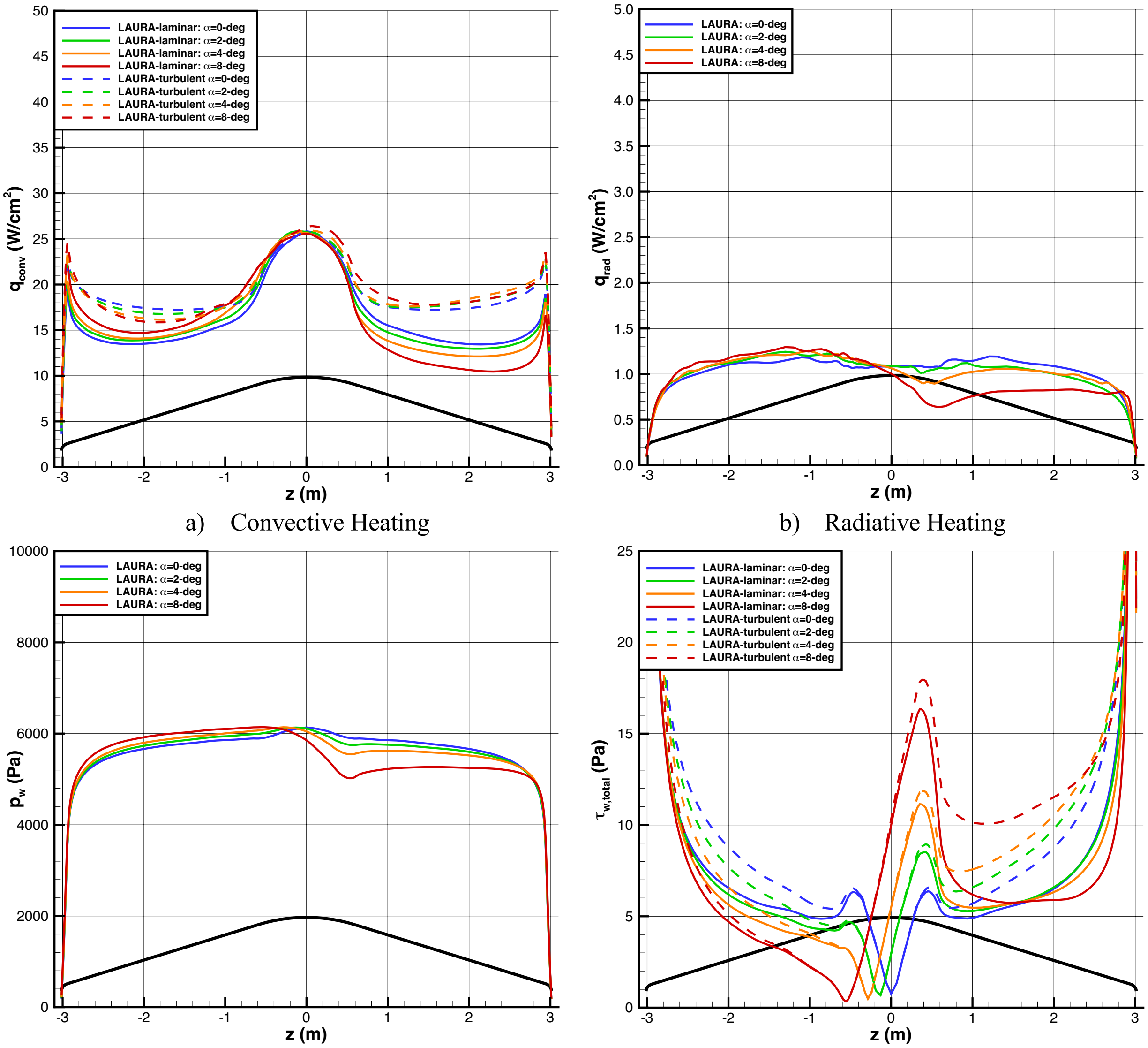


c) Surface Pressure



d) Surface Shear

Figure 13. Angle-of-attack effects on aerothermodynamic parameters, t = 488 s (rarefied/continuum overlap).



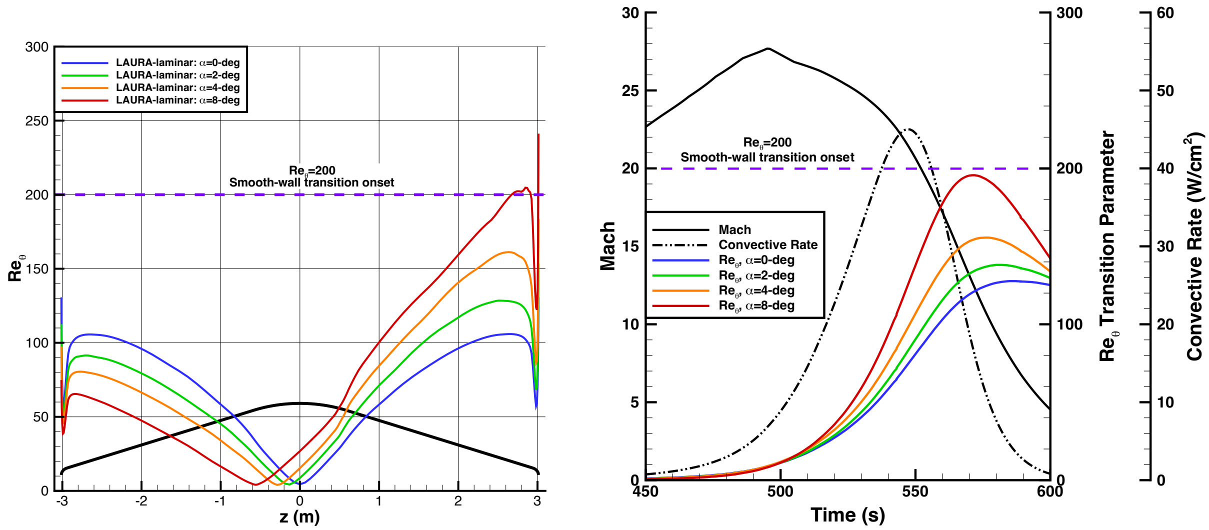
a) Convective Heating

b) Radiative Heating

c) Surface Pressure

d) Surface Shear

Figure 14. Angle-of-attack effects on aerothermodynamic parameters, t = 563 s (peak dynamic pressure).



a) Reθ Distribution at max dynamic pressure b) Reθ at shoulder along trajectory

Figure 15. Boundary-layer transition metrics.

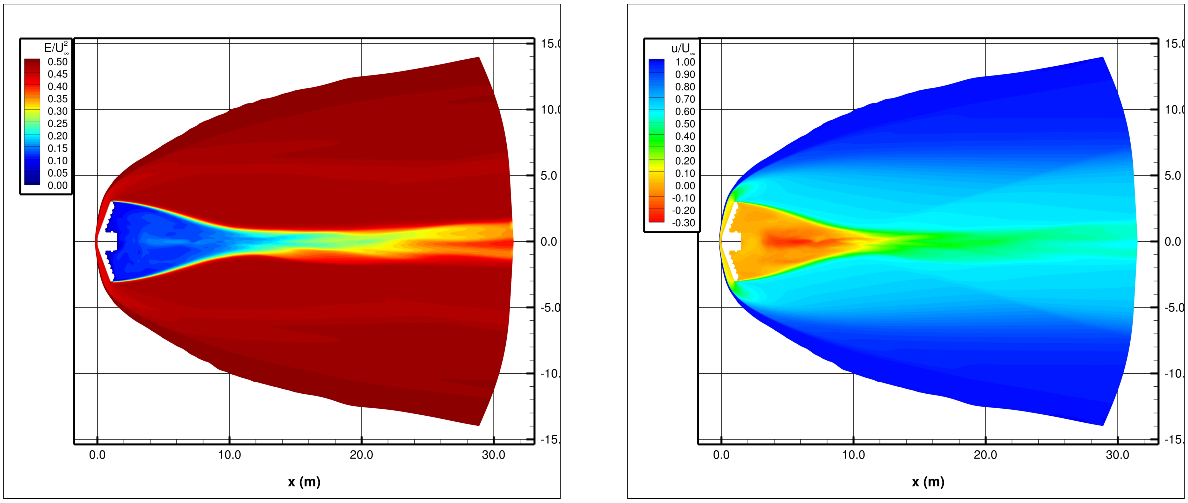
2. Aeroshell Back-Face and Payload Environments

Aerothermodynamic environments for the wake of the LOFTID aeroshell were based on laminar continuum LAURA simulations and MAP rarefied flow simulations at the EC_BN30_G13 trajectory points listed in Table 1. The LOFTID wake flow was defined by a large region of separated, low-energy flow with recirculating vortices (e.g. Figure 16 Several iterations of the database, each incorporating additional numerical complexity and improved grid generation processes were performed. The initial back-face database was generated using steady-state solutions on axisymmetric, conventional block-topology grids, but successive updates incorporated first-order time accuracy; a three-dimensional flow field domain; and the introduction and refinement of stacked-block topology grids. The improvements in the fidelity of wake flow simulations had a positive impact on the environment definition by producing consistently lower back-face heating estimates with each database update, as shown in Figure 17. Note that the stacked-block topology was only required for structured grid LAURA solutions; for MAP simulations the cartesian / cut-cell grid paradigm was amenable to the definition of complex surface features.

Owing to the complexity of producing structured grids for the complex topology of the exposed toroids on the back-face of the vehicle, the flight database heating environments were generated using a smooth (no exposed toroids) aeroshell back-face geometry. Over the course of the project, the stacked-block grid topology paradigm was developed and refined, eventually allowing for LAURA simulations of the actual vehicle geometry with exposed toroids. However, for consistency between updates of the database, only the smooth back-face grid solutions were incorporated into the database. Based on heating predictions from MAP, which did incorporate the toroids but were only available for high-altitude cases, as well as the wind tunnel aeroheating data discussed below, a multiplier factor of 1.25 was applied to the LAURA predictions to account for higher heating at the toroid heating peaks compared to that of a smooth OML. Examples of MAP and LAURA solutions over the OML with toroids are show in Figure 18.

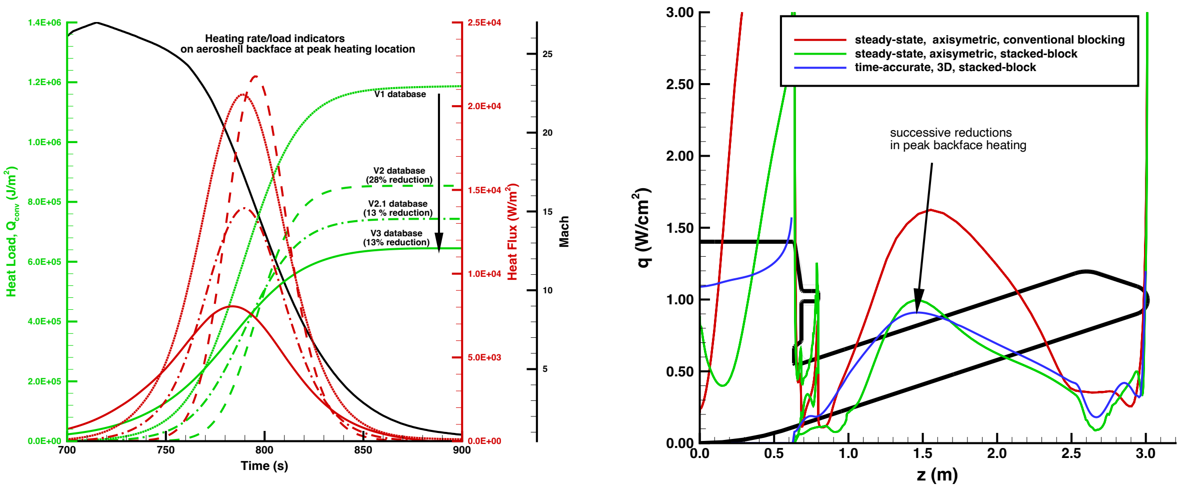
Convective heating and surface pressure distributions (radiative heating was negligible) on the back-face of the aeroshell are shown in Figure 19 for three separate times spanning the rarefied, high hypersonic, and low hypersonic to supersonic portions of the trajectory. Distributions along the payload for the same three times are shown in Figure 20. Note that the OML definition of the centerbody payload used for the MAP solutions was slightly different than that used for LAURA as the geometric features changed over the course of the project.

Good agreement between LAURA and MAP predictions was achieved in the overlap region where both solvers were employed. A noticeable change in the location of peak heating on the back-face of the aeroshell was noted between t = 753 s and t = 758 s. This region corresponded to a change in the character of the flow field solutions from relatively steady at higher altitudes to increasing unsteady and chaotic at lower altitudes.



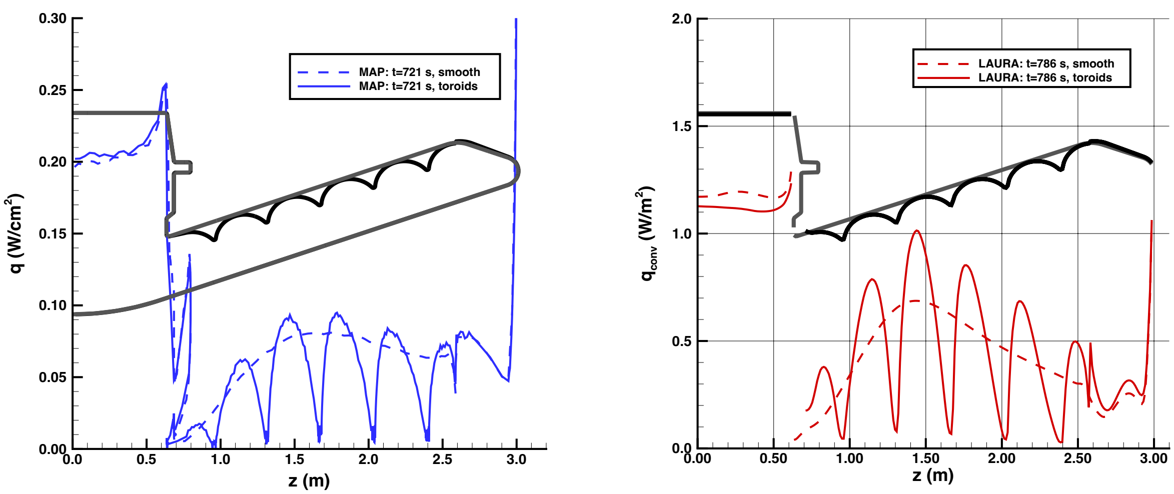
a) Total energy b) Velocity x-component

Figure 16. LOFTID wake flow.



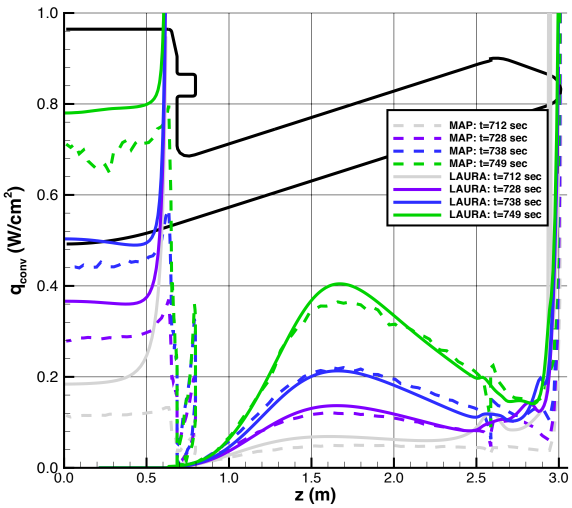
a) Time History b) Distribution at t = 776 s

Figure 17. Reductions in heating environments due to improvements in wake flow simulation methodology.

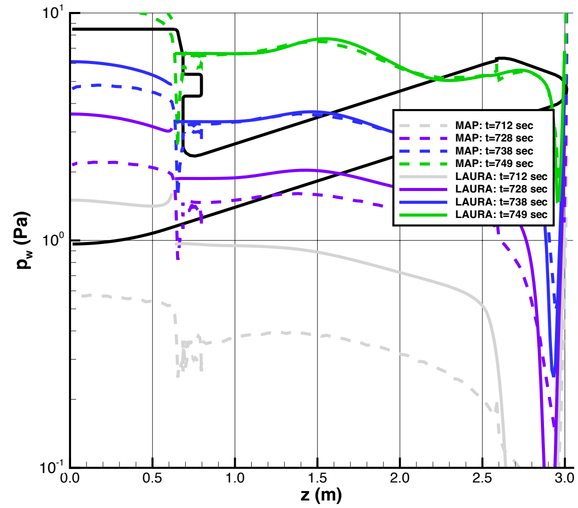


a) Map solutions b) LAURA solutions

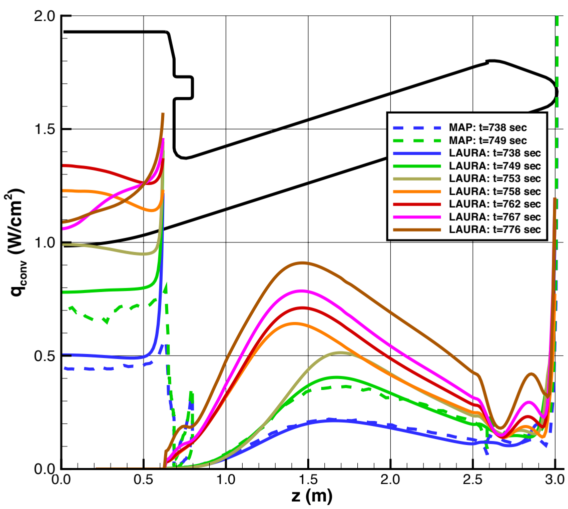
Figure 18. Comparison of smooth and toroid OML solutions.



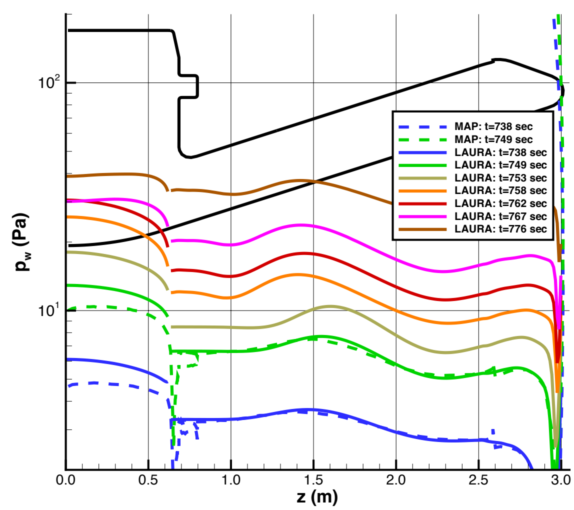
a) Convective heating, t=712 s to 749 s



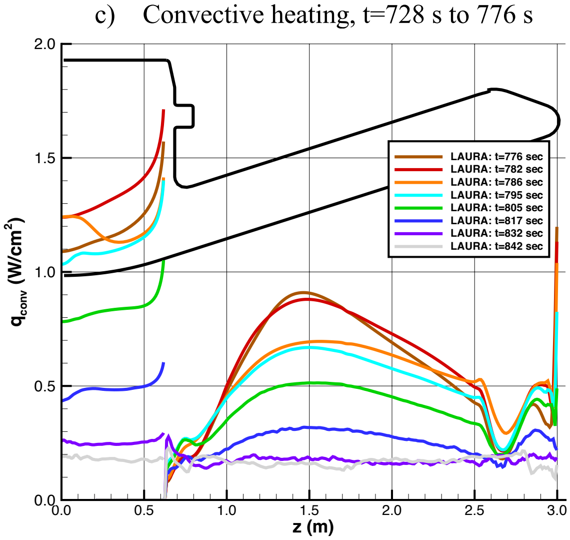
b) Pressure, t=712 s to 749 s



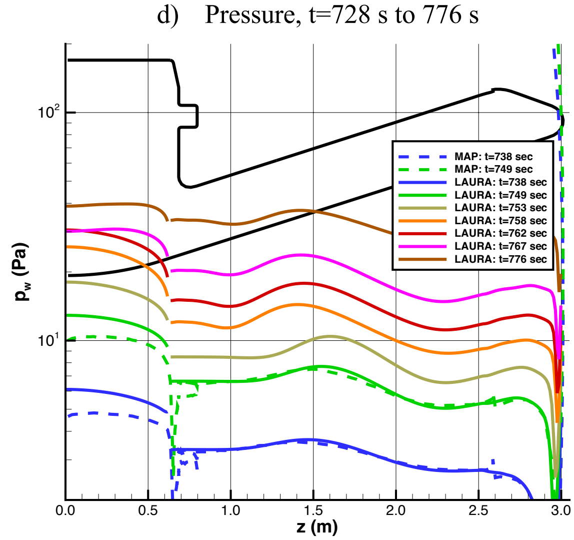
c) Convective heating, t=728 s to 776 s



d) Pressure, t=728 s to 776 s

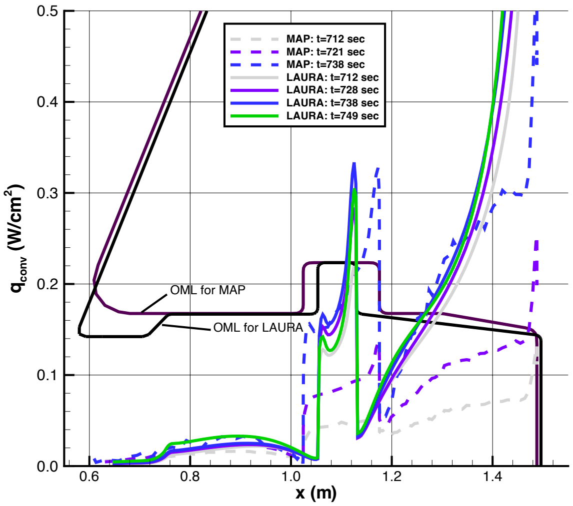


e) Convective heating, t=776 s to 842 s

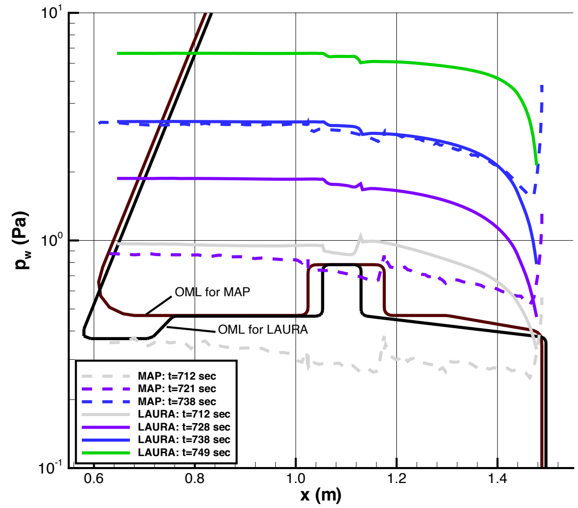


f) Pressure, t=776 s to 842 s

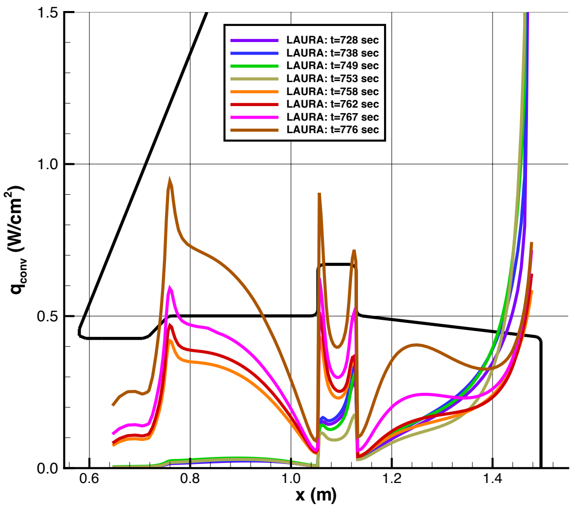
Figure 19. Aeroshell backface convective heating and pressure.



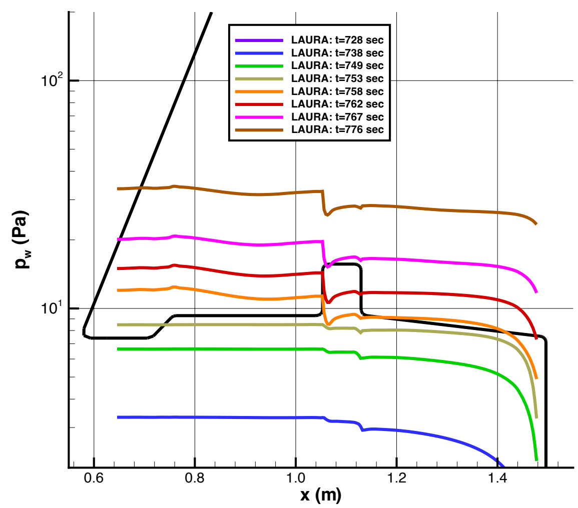
a) Convective heating, t=712 s to 749 s



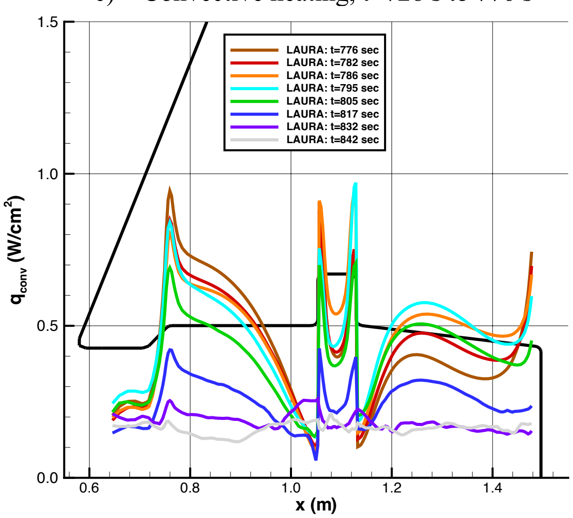
b) Pressure, t=712 s to 749 s



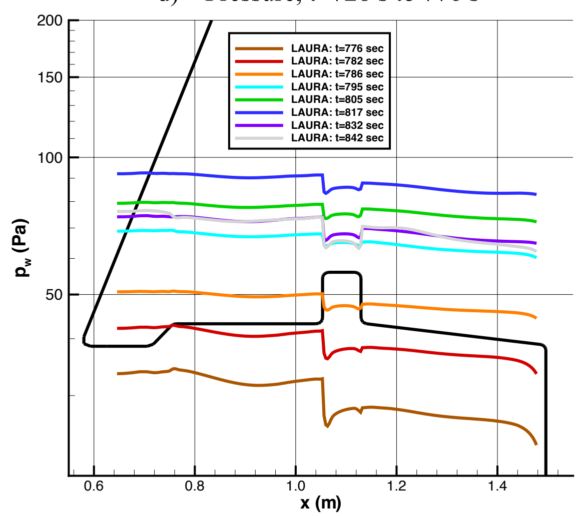
c) Convective heating, t=728 s to 776 s



d) Pressure, t=728 s to 776 s



e) Convective heating, t=776 s to 842 s



f) Pressure, t=776 s to 842 s

Figure 20. Centerbody convective heating and pressure.

C. Wind Tunnel Aeroheating and Boundary-Layer Transition Data

Two wind tunnel test programs were conducted to obtain data on the LOFTID vehicle for use in validation and uncertainty estimation of aeroheating predictions as well as to assess the effects of FTPS scalloping on boundary-layer transition. Aeroheating and boundary-layer transition data on the LOFTID forebody were obtained in the 20-Inch Mach 6 Air Tunnel. Because heating levels (and the associated temperature rise) for separated wake flows are an order of magnitude lower than on the forebody (e.g., Ref [34]), wake heating measurements are very challenging. Thus, the 31-Inch Mach 10 Air Tunnel, which has a much higher driver enthalpy than the 20-Inch Mach 6 Air Tunnel and thus produces higher heating, was used to obtain backface and centerbody heating data. Nominal conditions for these two test programs are tabulated in Table 3 for Test 7052 in the 20-Inch Mach 6 Air Tunnel and in Table 4 for Test 539 in the 31-Inch Mach 10 Air Tunnel.

Table 3. Conditions for 20-Inch Mach 6 Air Tunnel Test 7052.

Re_∞ (1/ft)	Re_∞ (1/m)	M_∞	T_∞ (K)	ρ_∞ (kg/m^3)	U_∞ (m/s)	ΔH (J/kg)	h_{FR} (kg/m^2-s)
1.049E+06	3.443E+06	5.89	62.35	1.652E-02	932.0	1.960E+05	1.662E-01
2.037E+06	6.683E+06	5.97	62.80	3.180E-02	947.0	2.105E+05	2.353E-01
3.018E+06	9.904E+06	6.00	63.03	4.700E-02	953.0	2.165E+05	2.882E-01
4.996E+06	1.639E+07	6.03	63.58	7.782E-02	961.4	2.250E+05	3.748E-01
6.522E+06	2.140E+07	6.04	63.34	1.013E-01	960.7	2.241E+05	4.271E-01
7.255E+06	2.380E+07	6.05	63.29	1.125E-01	960.8	2.241E+05	4.503E-01
8.175E+06	2.682E+07	6.03	59.15	1.229E-01	924.2	1.853E+05	4.486E-01

Table 4. Conditions for 31-Inch Mach 10 Air Tunnel Test 539.

Re_∞ (1/ft)	Re_∞ (1/m)	M_∞	T_∞ (K)	ρ_∞ (kg/m^3)	U_∞ (m/s)	ΔH (J/kg)	h_{FR} (kg/m^2-s)
5.301E+05	1.739E+06	9.68	51.86	4.607E-03	1397.5	7.277E+05	1.424E-01
1.048E+06	3.440E+06	9.81	50.99	8.920E-03	1403.2	7.348E+05	1.991E-01
1.862E+06	6.109E+06	9.94	49.66	1.546E-02	1400.4	7.294E+05	2.615E-01

1. Aeroshell Front-Face Aeroheating and Boundary-Layer Transition

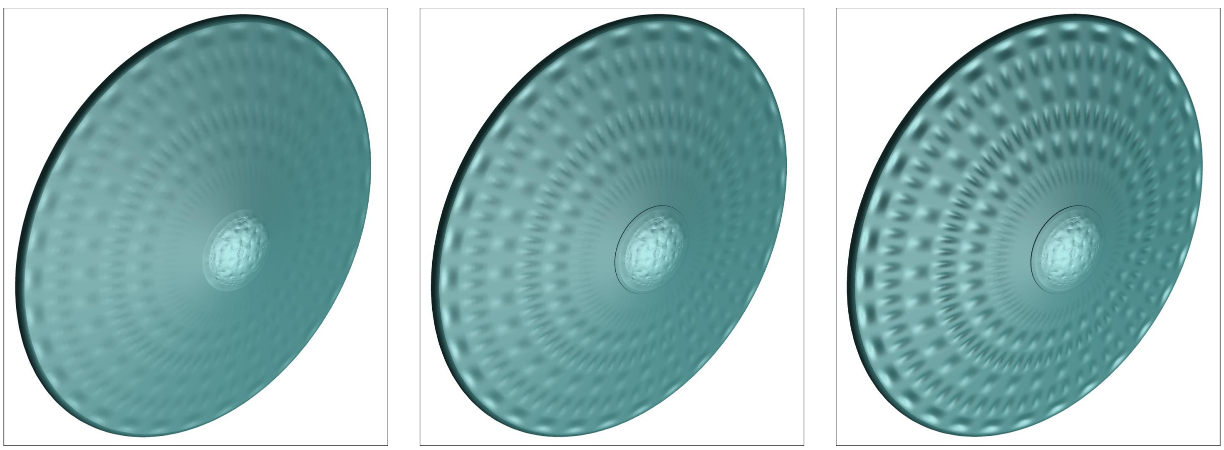
The OML of an inflatable aeroshell like that of LOFTID is subject to deformation under aerodynamic loading as the flexible thermal protection system material deforms over the underlying rigid toroids and the tension-tie straps that hold them together, which is referred to as scalloping. Any deviations from a smooth OML, such as scalloping, have the potential to promote early boundary-layer transition and turbulent heating augmentation depending on the physical scale of the feature with respect to the boundary-layer height. Because there are no reliable computational methods to predict transition due to scalloping, experimental data were acquired to assess the effects of scalloping on the LOFTID aeroshell. Scalloping effects data have been generated for the IRVE-3 geometry [35], but because the LOFTID geometry has a larger cone angle and much larger nose radius, additional data were required.

Scalloping effects for the LOFTID geometry were studied through aeroheating testing in the LaRC 20-Inch Mach 6 Air Tunnel using global phosphor thermography. The wide Reynolds number range of this tunnel allowed for testing of models across a range of conditions that produced laminar, transitional, and turbulent boundary-layer flow over the aeroshell. Data were obtained on 6-inch (0.1524 m) diameter cast ceramic wind tunnel models of the LOFTID aeroshell with different levels of FTPS scalloping at seven unit Reynolds number conditions from $Re_\infty = 1.0\times10^6$/ft to 8.2×10^6/ft at angles of attack of $\alpha = 0$ deg, 4 deg and 8 deg.

Data on FTPS response to structural loading obtained through testing [36–37] in the National Full-Scale Aerodynamic Complex (NFAC) performed in support of the IRVE-3 program were used to provide estimates for the expected in-flight scalloping for LOFTID. The expected worst-case deformation at the peak dynamic pressure condition was used to define a baseline geometry that had 0.24 mm scallop height when scaled to wind tunnel model size. Two additional wind tunnel models with scallop heights of half (0.12 mm) and twice (0.48 mm) the baseline height were also designed to obtain data on the effects of scallop height variation on transition. Renderings of these models are shown in Figure 21. Notably, these models all incorporated the backward-facing step between the solid nose cap and inflatable aeroshell, as well as the "quilting" of the FTPS covering the nose that were both part of the actual flight vehicle.

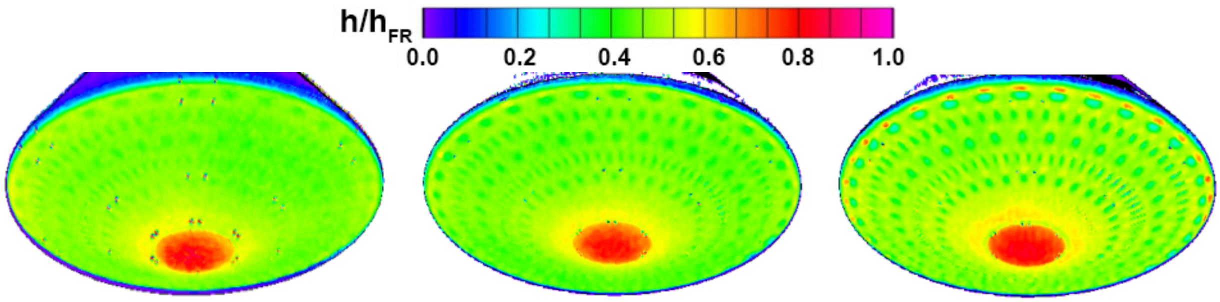
Sample aeroheating distributions for each of these geometries across the range of Reynolds numbers at $\alpha = 0$ deg are presented in Figure 22. The flow over the small scallop model remained laminar at all conditions. However, the nominal scallop model flow became transitional at $Re_\infty = 7.3\times10^6$/ft and the large scallop model flow was transitional at $Re_\infty = 5.0\times10^6$/ft and fully turbulent at $Re_\infty = 8.2\times10^6$/ft. Data from the higher angle-of-attack cases (not shown) indicated leeside transition at lower Reynolds numbers as would be expected. The influence of the backward facing step at the nose / cone junction can also be seen in the sharp dip in heating immediately downstream of the nose.

Although the large-scallop, high Reynolds number data showed dramatic heating increases, the overall assessment of this data set was that scalloping effects would have minimal impact at the LOFTID flight conditions. As shown previously in Figure 15, predicted flight Re_θ values remain below the smooth-wall transition onset value of 200 throughout the trajectory, even for high angle-of-attack excursion from the nominal 0 deg angle of attack. While scalloping can produce transition before the smooth wall limit is reached, the wind tunnel data indicated that significant boundary-layer transition would not occur at LOFTID flight conditions. A comparison of the flight Re_θ levels for the peak convective heating and peak dynamic pressure trajectory points to the those at the wind tunnel test conditions is shown in Figure 23. From this figure, it can be concluded that only the wind tunnel data for $Re_\infty < 5.0\times10^6$/ft are relevant to flight conditions. At these conditions, only minimal scalloping effects were observed, even on the large scallop depth model. As shown in Figure 24, these small effects are bounded by fully-turbulent heating predictions, which were used to define the flight aeroheating database.

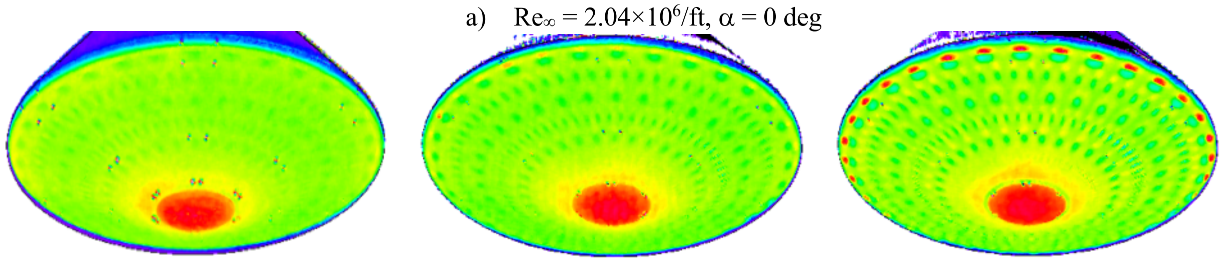


a) Small scallop depth b) Nominal scallop depth c) Large scallop depth

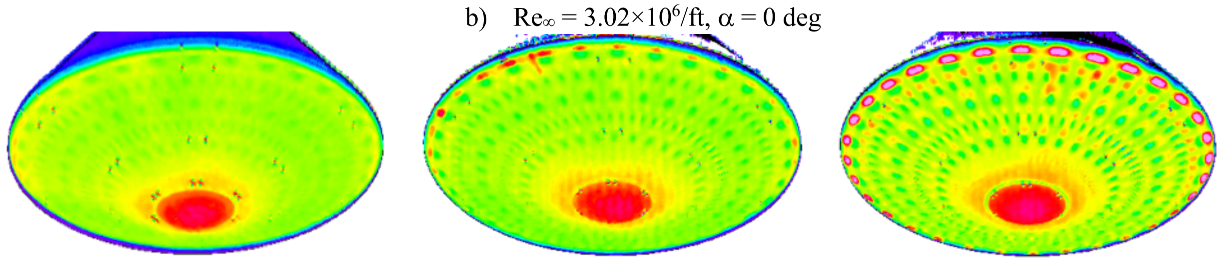
Figure 21. CAD renderings of scalloped OML wind tunnel models.



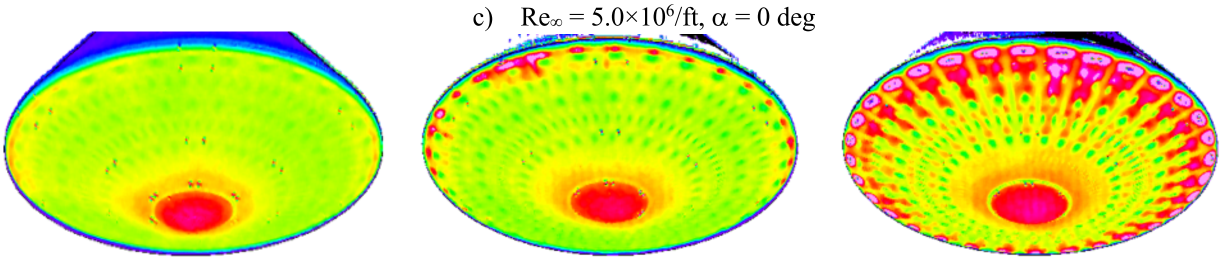
a) $Re_\infty = 2.04\times10^6$/ft, $\alpha = 0$ deg



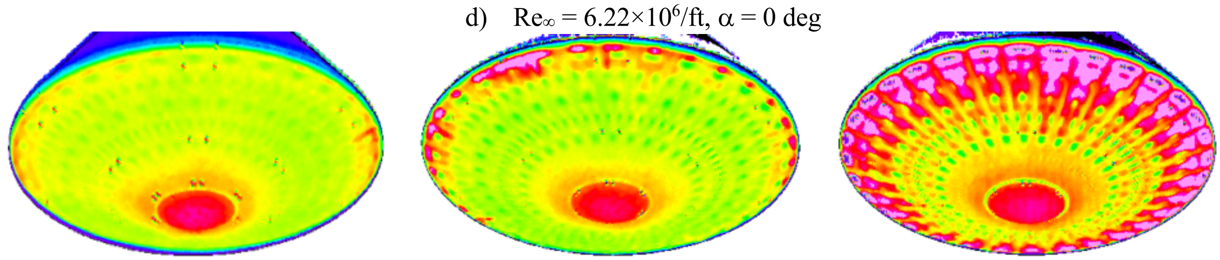
b) $Re_\infty = 3.02\times10^6$/ft, $\alpha = 0$ deg



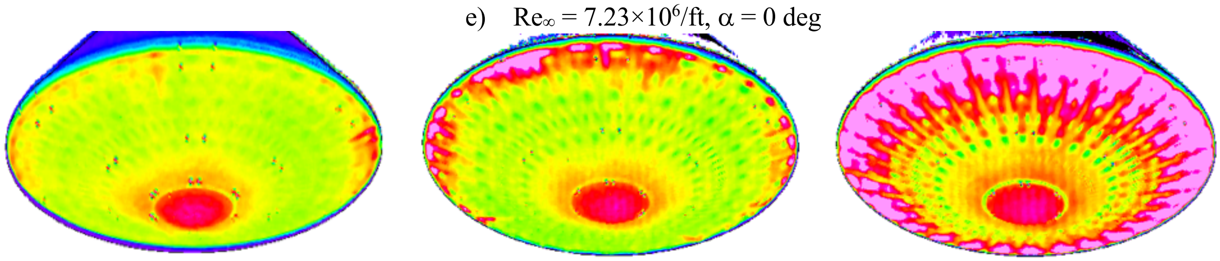
c) $Re_\infty = 5.0\times10^6$/ft, $\alpha = 0$ deg



d) $Re_\infty = 6.22\times10^6$/ft, $\alpha = 0$ deg

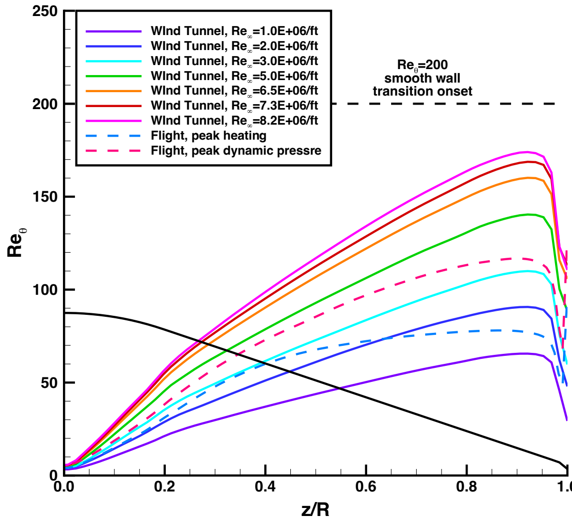


e) $Re_\infty = 7.23\times10^6$/ft, $\alpha = 0$ deg



f) $Re_\infty = 8.18\times10^6$/ft, $\alpha = 0$ deg

Figure 22. FTPS scalloping effects on heating and transition (small, nominal, and large depths).



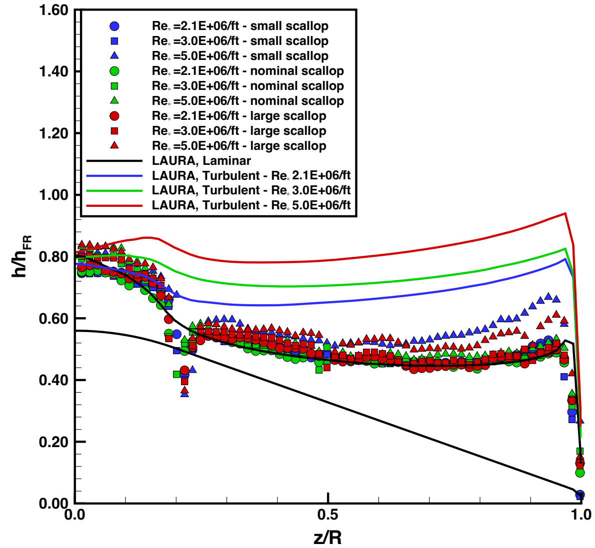
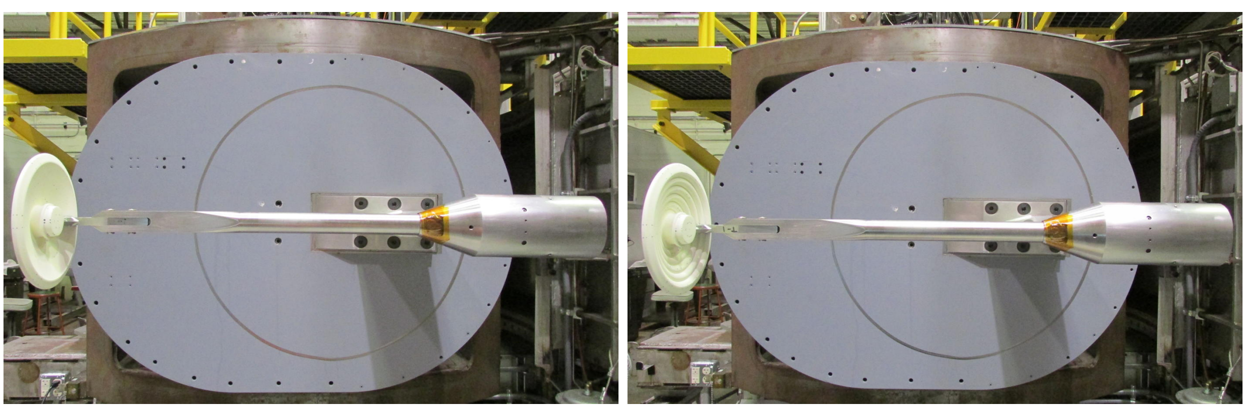


Figure 23. Comparison of flight and wind tunnel Re_θ values at α = 0 deg.

Figure 24. Comparison of scallop heating data to predictions at α = 4 deg.

2. Aeroshell Back-Face and Payload Heating

Because of the modeling challenges for the separated, unsteady flow in the wake of the LOFTID vehicle, experimental data were required for validation of the computational methodology. These data were obtained through aeroheating testing in the LaRC 31-Inch Mach 10 Air Tunnel. Testing was performed at three Reynolds numbers $Re_\infty = 0.53\times10^6$/ft to 1.8×10^6/ft with an angle-of-attack range of +4 to -4 deg. Global phosphor thermography aeroheating measurements were made on 6-inch (0.1524 m) diameter cast ceramic models of LOFTID with both a smooth backface geometry and with the exposed toroids. As shown in Figure 25, these models were mounted on a narrow blade fixture rather than a circular sting to minimize support hardware interference with the wake flow and allow data to be obtained on the centerbody payload; however, this mounting arrangement did produce some flow asymmetry.



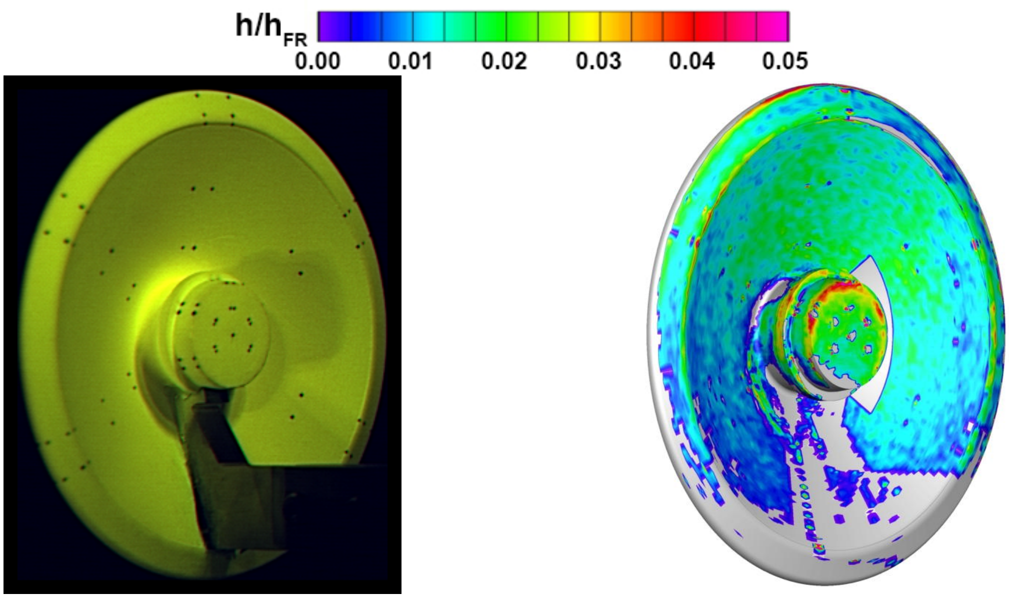
a) Smooth backface model

b) Toroid backface model

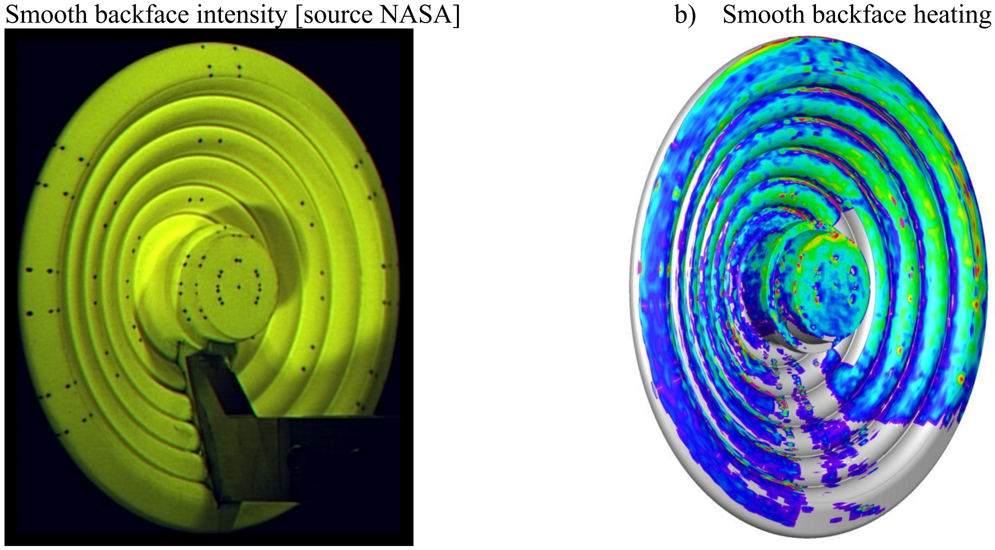
Figure 25. LOFTID models in 31-Inch Mach 10 Air Tunnel [source NASA].

Sample measured intensity and processed heating results at the highest Reynolds number condition are shown in Figure 26(a) for the smooth model and Figure 26(b) for the toroid model. Even with the higher driver enthalpy of this tunnel, the surface temperature rise on the backface of the model was less than 5K, leading to considerable scatter in the measurements over the model surface. To reduce the experimental uncertainty, data were extracted along multiple, circumferentially-spaced rays and averaged as a function of radial position. These results are plotted in Figure 27(a) for the smooth model and Figure 27(b) for the toroid model. The averaged surface temperature and uncertainty bars for the heating are also shown in these figures.

Comparisons between these data and predictions generated using the LAURA code are shown in Figure 28(a)–(b) for the smooth model and in Figure 28(c)–(d) for the toroid model. These predictions were generated using three-dimensional, time-averaged, unsteady simulation on stacked-block topologies as described in the Computational Grid Topologies section. While the comparisons for the more complicated toroid topology were considerably better than those for the smooth topology, these results are somewhat misleading. The smooth-model simulations were the first performed using the new stacked-block paradigm and the resulting grid quality was less than ideal. Over multiple iterations of the flight database, grid quality improvements were realized as more experience was gained generating stacked-block grids, leading to a better grid and more accurate comparisons for the toroid topology model. These comparisons for the toroid cases provided confidence in application of the stacked block topology to flight conditions. Because of the high CPU usage required to generate these unsteady solutions, further attempts to simulate the smooth back-face topology were deferred in favor of computations supporting the flight database.

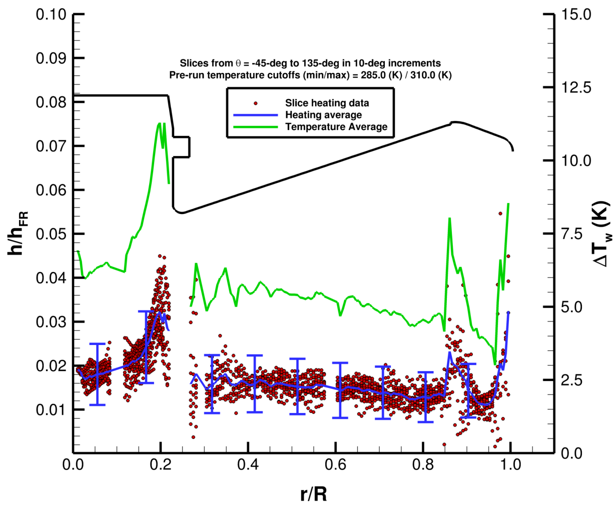


a) Smooth backface intensity [source NASA] b) Smooth backface heating

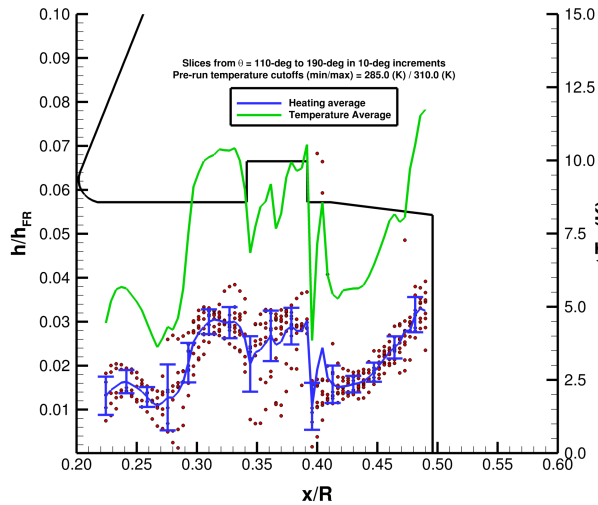


c) Toroid backface intensity [source NASA] d) Toroid backface heating

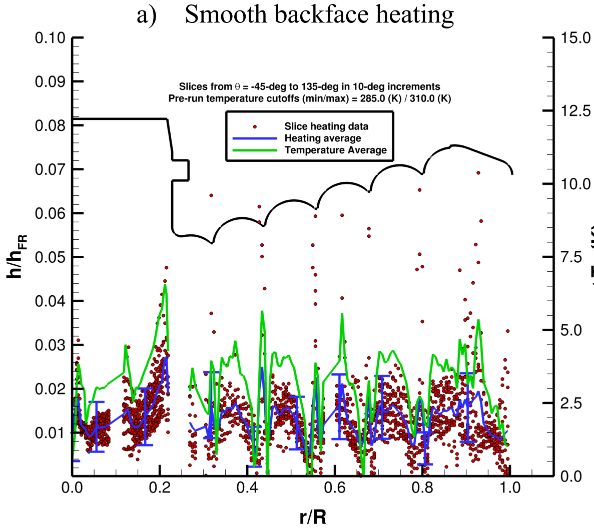
Figure 26. Intensity and heating data from 31-Inch Mach 10 Air Tunnel, $Re_\infty = 1.9\times10^6$/ft, α = 0 deg.



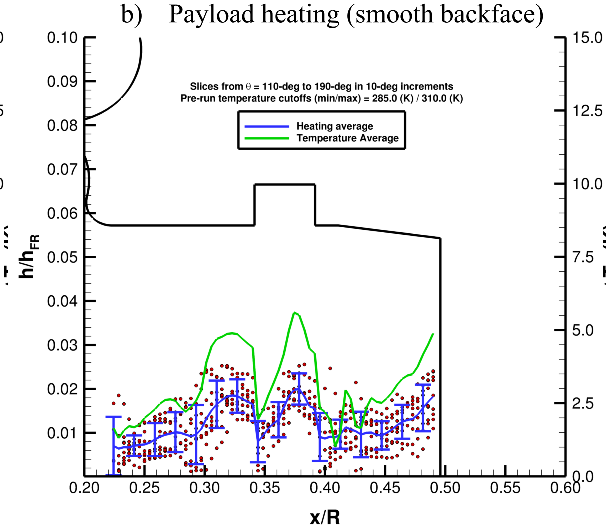
a) Smooth backface heating



b) Payload heating (smooth backface)

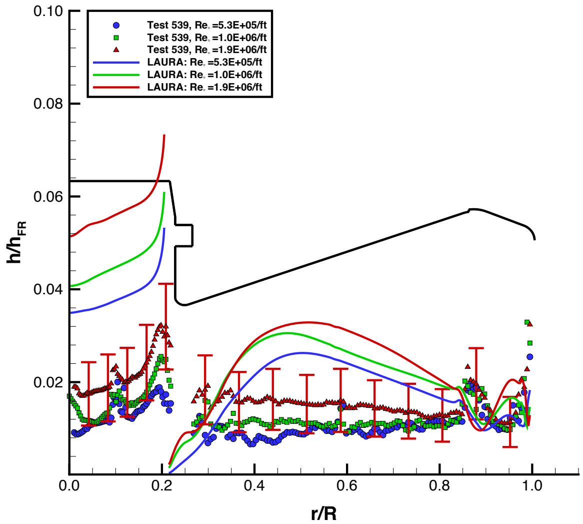


c) Toroid backface heating

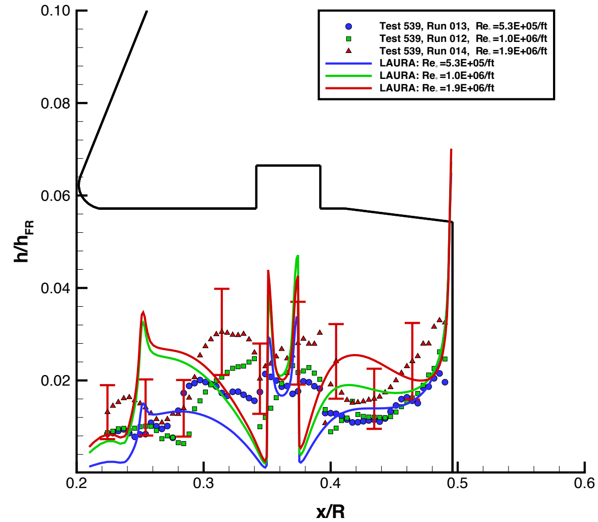


d) Payload heating (toroid backface)

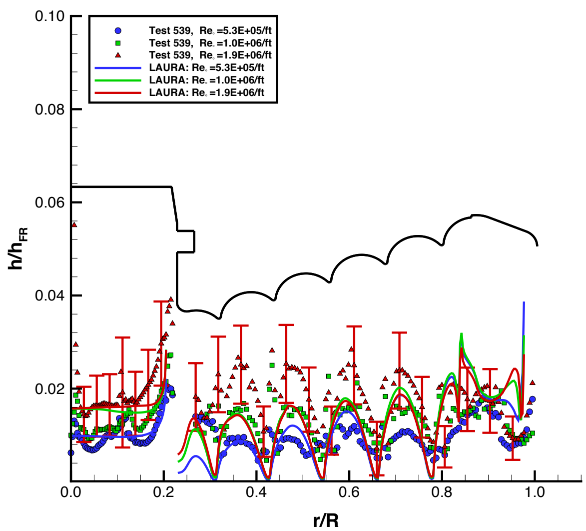
Figure 27. Distributed and averaged data from 31-Inch Mach 10 Air Tunnel, $Re_\infty = 1.9\times10^6$/ft, $\alpha = 0$ deg.



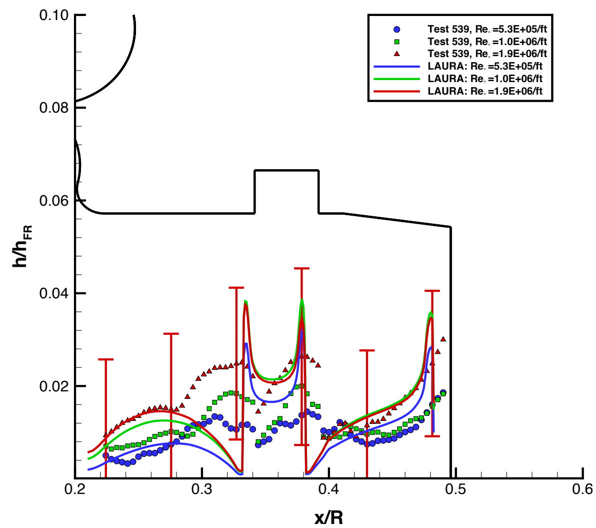
a) Smooth backface



b) Payload (smooth backface)



c) Toroid backface



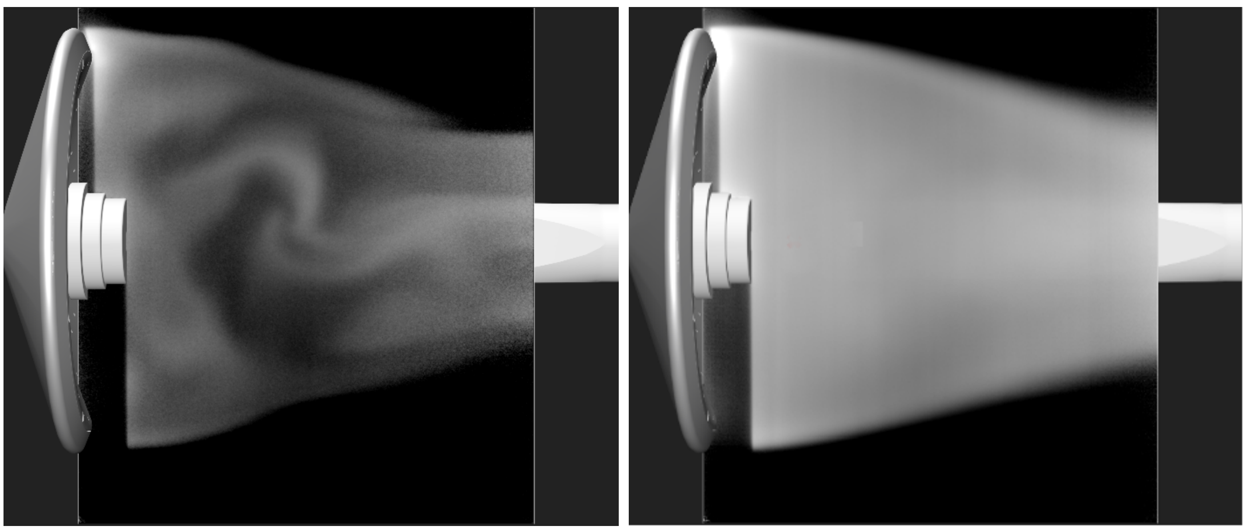
d) Payload (toroid backface)

Figure 28. Comparison of 31-Inch Mach 10 Air Tunnel data with LAURA predictions.

D. Wind Tunnel Flow Field Visualization Data

Flow field visualization studies [36] were performed during two test entries in the LaRC 31-Inch Mach 10 Air Tunnel using both NO-PLIF and FLEET measurement techniques to obtain velocimetry data. This effort was supported jointly by the NASA Entry Systems Modeling project (ESM) and the LOFTID program. The focus of this work was to demonstrate the ability to obtain quantitative flow field data for computational fluid dynamics validation. The ESM project was the initial customer for these data as the test program was conducted after the LOFTID design work was completed. However, the data from these studies, as well as planned follow-on studies, will be used to support a detailed, post-flight reconstruction of the LOFTID vehicle performance.

Example NO-PLIF results from a $Re_\infty = 1.8\times10^6$/ft case are shown in Figure 29. These figures are the instantaneous and time-averaged PLIF signal intensity, which are useful for flow visualization purposes. The free shear-layer that defines the boundary of the recirculating flow regions immediately behind the vehicle is clearly delineated in both images, and the instantaneous image demonstrates that the recirculation region flow field is chaotic and unsteady. Velocity measurements using the FLEET technique are shown in Figure 30. These data represent 10,000 instantaneous velocity measurements acquired at a repetition-rate of 1 kHz at a point on the centerline of the wake flow behind the model.



e) Instantaneous f) Time-averaged

Figure 29. NO-PLIF signal intensity visualization of LOFTID wake flow structure.

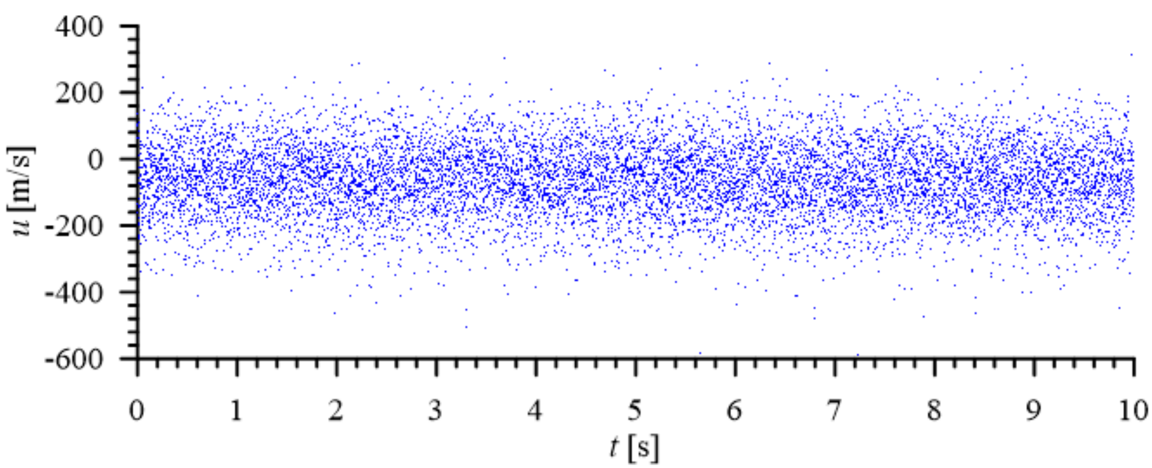


Figure 30. FLEET measurements of instantaneous wake centerline velocity.

V. Summary

Aerothermodynamic databases for the LOFTID flight test demonstration mission were developed using a combination of computational and experiment methods. The MAP, FUN3D, LAURA, and HARA codes were used to provide aerodynamic forces and moments and convective and radiative heating rates and loads to define the expected flight performance of the vehicle. Experimental ground testing was conducted in the NASA LaRC 31-Inch Mach 10 Air Tunnel and 20-Inch Mach 6 Air Tunnel to obtain data on flexible thermal protection system deflection effects on boundary-layer transition and turbulent heating augmentation, as well as to provide wake flow data for comparison with computational results to help define predictive uncertainties. These databases provided the aerodynamic information required for trajectory simulations of the vehicle flight path and the aeroheating information required for thermal analyses of the thermal protection system. The LOFTID flight successfully executed in November of 2023, with the vehicle meeting all performance metrics to demonstrate the viability of inflatable aeroshell technology at mission relevant conditions.

VI. Acknowledgements

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